Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Inference

Daniel Zhao

Supervisor: Associate Professor Joseph T. Lizier

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School of Computer Science
The University of Sydney
Australia

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Abstract

Mutual information (MI) measures association between two variables. This thesis compares the MI of two independently sampled categorical populations, allowing their joint probability tables to differ even when their MI values are equal. The Normal Wald test uses a normal approximation that can be inaccurate in small or sparse samples.

The thesis develops an Expanded Welch–Satterthwaite adjustment by deriving how changing one observation affects the estimated MI variance. The calculation includes changes to cell probabilities and their row and column totals. It supplies the degrees of freedom for a Student t distribution, while retaining Wald's estimated difference and standard error.

Three controlled studies examine false positives, association strength and detection. The design uses $2 \times 2, 4 \times 4, 8 \times 8$ and 16×16 probability tables, three margin regimes and equal sample sizes, with 20,000 sampled table pairs in each of 232 settings. Bias and variance diagnostics explain the observed patterns, and a small alternative-construction check assesses a representative comparison.

Expanded Welch lowers rejection, helping where Wald rejects equal MI too often but making conservative rates lower still. Near independence, small estimated degrees of freedom reinforce this conservatism. In 16×16 different-skew tables at $n = 1{,}000$, unequal MI bias shifts the estimated difference away from zero: false-positive rates are 0.2522 for Wald and 0.2402 for Expanded Welch despite full validity. At $n = 10{,}000$, the bias is much smaller and the rates are 0.0463 and 0.0453. Larger samples also improve detection and reduce the difference between methods.

Changing the reference distribution cannot correct errors in the shared MI difference or standard error. Normal Wald remains the simpler baseline for this task, with Expanded Welch a potential supplementary check where a closely matched calibration study supports its use.

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Contents

Abstract i

Acknowledgements ii

1 Introduction 1

1.1 Why the comparison is difficult . . . 2
1.2 Research questions . . . 2
1.3 Contributions to the field . . . 3
1.4 Scope and organisation . . . 3

2 Background and Literature Review 4

2.1 Information-theoretic foundations . . . 4
2.1.1 Entropy and information content . . . 4
2.1.2 Pointwise and average mutual information . . . 4
2.2 Estimating discrete mutual information . . . 5
2.3 Sampling variance of plug-in MI . . . 7
2.4 Approaches to Testing MI Differences . . . 11
2.5 Previous work and this thesis's contribution . . . 14

3 The Two-Sample MI Test and Wald Baseline 16

3.1 Problem setup and bias-corrected estimator . . . 16
3.2 Shared standard error and test statistic . . . 18
3.3 Normal Wald reference and motivation for expansion . . . 18

4 The Expanded Welch–Satterthwaite Test for MI 20

4.1 Choosing Degrees of Freedom for Expanded Welch . . . 20
4.2 How the MI Variance Changes with the Table . . . 22

4.3 Component degrees of freedom and final test 25
4.4 Worked example and implementation 28
4.5 Behaviour and limitations 30

5 Experimental Design 32
5.1 Questions and controlled comparisons 32
5.2 Constructing joint distributions with known MI 33
5.3 The three studies and construction check 35
5.3.1 Sample size, table size and margins 35
5.3.2 Association strength and degrees of freedom 35
5.3.3 Detection of a true MI difference 36
5.3.4 A supporting change in construction 36
5.4 Sampling and performance measures 37

6 Experimental Results and Discussion 39
6.1 False positives across sample size and population structure 39
6.1.1 Increasing table size 39
6.1.2 Changing the margin regime 41
6.2 Association strength and degrees of freedom 43
6.3 Detection when population MI differs 45
6.3.1 Detection across table sizes 45
6.3.2 Detection across margin regimes 47
6.4 Computation and practical implications 49

7 Implications and Limitations 50
7.1 Implications for using Expanded Welch 50
7.2 Limits of the evidence 51

8 Conclusion 53

A Supporting Calculations 55
A.1 Mean of the variance influence function 55
A.2 Fixed-margin behaviour near independence 55
A.3 Dependence between the MI and variance estimates 57
A.4 The independence boundary 58

A.5 The combined formula when a component is zero 59

B Experimental Details **61**

B.1 Copying construction and target selection 61

B.2 Four-cell supporting construction 62

B.3 Complete parameter grids and sampling 63

B.4 Validity and sparse-sample measures 64

B.5 Bias, variance and degrees-of-freedom diagnostics 64

B.6 Simulation uncertainty 65

C Additional Results **66**

C.1 Complete equal-MI results 66

C.2 Supporting estimation diagnostics 71

C.3 Detection over the full difference range 72

C.4 Four-cell construction check 74

List of Figures

6.1 False positives and validity under different-skew margins. 40
6.2 False positives and validity across three margin regimes. 41
6.3 Association strength, false positives and estimated degrees of freedom. 44
6.4 Detection over small MI differences under different-skew margins. . 46
6.5 Detection over small MI differences in 4×4 tables. 48

C.1 Additional equal-MI comparisons under uniform and same-skew margins. 70
C.2 Full-range detection under different-skew margins. 73
C.3 Full-range detection across 4×4 margin regimes. 74

List of Tables

4.1 Operational Expanded Welch–Satterthwaite algorithm. 29

5.1 The three controlled studies and supporting construction check. . . 33
5.2 Row and column margins for each population. 34

6.1 Bias and uncertainty of the estimated MI difference. 42
6.2 Time per test call in microseconds: median [interquartile range]. . . 49

B.1 Exact parameter sets; MI values are in nats. 63

C.1 False-positive and valid-result rates for all 120 null settings. 66
C.2 Diagnostics for 4×4 different-skew populations with equal MI. . . . 71
C.3 Different-skew null diagnostics at $n = 50{,}000$. 72
C.4 Rejection and difference bias under two population constructions. . 75

Chapter 1

Introduction

Suppose two independent groups are observed on the same categorical variables X and Y. Mutual information (MI) describes the strength of association between those variables in each group. If the estimated MI differs, the difference alone does not tell us whether the population MI values differ: both estimates vary when new observations are drawn. The question in this thesis is whether the two *population* MI values can plausibly be equal.

For example, suppose X is a categorical input feature and Y is a class label in two independently collected datasets. The feature may have a higher estimated MI with the label in one dataset than in the other. Before interpreting this as a difference in association, we need to distinguish a population difference from variation caused by sampling. The datasets can have different feature and class frequencies, so the question is not whether all their cell probabilities match. It is whether the feature–label association, as measured by MI, is equally strong in the two populations.

This is not the usual test of whether X and Y are independent within one group. The null hypothesis here permits dependence in both groups and permits their complete joint distributions to differ. It specifies only

$$H_0 : I(P) = I(Q), \qquad H_1 : I(P) \neq I(Q), \tag{1.1}$$

where P and Q are the two joint distributions. The samples from P and Q are independent of each other; that sampling assumption does not imply independence of X and Y within either population.

1.1 Why the comparison is difficult

MI uses the probabilities in every cell of a table and the totals of its rows and columns. In a small sample, some cells can be empty even when their population probabilities are positive. The observed tables determine both the estimated MI difference and its standard error, so sparse data can affect both. The Normal Wald test compares their ratio with a standard normal distribution, but in small samples it may reject equal MI more or less often than the intended 5%.

Expanded Welch offers a possible adjustment. It uses the same estimated difference and standard error as Wald, but compares their ratio with a Student t distribution. Choosing that Student t distribution requires a calculation specific to MI. Its degrees of freedom depend on how much the estimated MI variance changes from sample to sample. To calculate that change, we must account for the fact that changing one cell also changes its row and column totals and the MI values calculated from them. This thesis derives that calculation and tests whether the adjustment helps.

Expanded Welch can reduce false positives where Wald rejects too often, but it can also reduce detection where Wald already rejects too rarely. In sparse samples, it may fail to return a usable p-value when Wald succeeds. The experiment therefore examines false positives, detection and the frequency of usable results together.

1.2 Research questions

The thesis asks two main questions. First, how can changes in the estimated MI variance be used to choose the degrees of freedom for a test using a Student t distribution?

Second, does the resulting Expanded Welch test offer a useful improvement over Normal Wald? The experiment checks how false positives, detection and validity change with sample size, table size, margin regime and association strength. Bias and variance diagnostics explain the main patterns, and a small alternative-construction check tests a representative comparison under a different association pattern. It also measures the extra computation required.

1.3 Contributions to the field

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated MI variance. The calculation includes the resulting changes in the row and column totals and in pointwise MI.

2. **Methodological contribution.** Expanded Welch uses this calculation to choose the degrees of freedom for a two-sample test using a Student t distribution.

3. **Empirical contribution.** A reproducible experiment measures false positives, detection, usable results and runtime on fixed populations with known MI. Three controlled studies cover 2×2 through 16×16 square tables, three margin regimes and positive MI targets at equal sample sizes. Estimation diagnostics connect observed test performance to errors in the shared difference and standard error.

4. **Practical contribution.** Expanded Welch can help when Wald rejects equal MI too often. It can also make an already-low rejection rate lower and return fewer usable results in sparse samples. Changing the reference distribution alone cannot correct errors in the shared MI difference or standard error.

1.4 Scope and organisation

The analysis concerns independent samples recorded in count tables with the same row and column categories. The main derivation assumes every population cell has positive probability and that its leading large-sample MI variance is greater than zero. Exact independence is considered separately because that variance becomes zero. Within this setting, the experiment compares Normal Wald with Expanded Welch using a two-sided 5% test. It does not establish performance for paired data, continuous MI estimators or a number of categories that grows with sample size.

Chapter 2 supplies the statistical background. Chapters 3 and 4 define the shared statistic and derive the correction. Chapter 5 explains how the populations and samples are generated and how the results are measured. Chapter 6 presents and interprets the main comparisons in sequence. Chapter 7 draws out their broader implications and limits, and Chapter 8 states the contribution and recommendation. The appendices give the detailed algebra, population settings, simulation diagnostics and supporting figures.

Chapter 2

Background and Literature Review

Testing whether two populations have equal MI requires an estimate from each sample and a measure of its sampling error. This chapter develops those inputs and reviews methods for comparing them.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = -\sum_x p_X(x) \log p_X(x). \tag{2.1}$$

Entropy summarises the uncertainty of one random variable before its value is observed (Shannon, 1948).

For a pair (X, Y), the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X \mid Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y.

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \ldots, r\}$ and $Y \in \{1, \ldots, c\}$. Write

$$p_{i+} = \sum_{j=1}^{c} p_{ij}, \qquad p_{+j} = \sum_{i=1}^{r} p_{ij} \tag{2.2}$$

for the row and column margins. The pointwise mutual information associated with cell (i, j) is

$$\ell_P(i, j) = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right). \tag{2.3}$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts; a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^{r}\sum_{j=1}^{c} p_{ij}\ell_P(i, j). \tag{2.4}$$

Here the summations are nested: for each row i, the inner sum runs over all columns j. Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$.

Because P is the joint distribution of (X, Y), we also write $I(P)$ as $I(X; Y)$. Its entropy form is

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \tag{2.5}$$

The quantity is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells (Bossomaier et al., 2016).

Although MI measures the overall departure from independence, it does not specify the shape of association. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. Our null requires only this one number to agree, not that the two tables match cell by cell.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by $\log 2$ converts to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij}. The empirical cell probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}}{n}. \tag{2.6}$$

Substitution into Equation (2.4) gives the plug-in estimator

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij} \log \left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}} \right). \tag{2.7}$$

The zero-cell convention follows from $\lim_{x \downarrow 0} x \log x = 0$. Empty cells contribute no direct term, but they still affect the empirical margins and the finite-sample behaviour of the estimator.

Under multinomial sampling, the cell counts vary between samples but must sum to n. An extra observation in one cell therefore leaves fewer observations for the others. For a fixed set of categories, the plug-in estimate approaches population MI as the sample grows, but its small-sample behaviour depends on the complete probability table. In particular, rare cells can change this behaviour even when the categories and sample size are unchanged (Paninski, 2003).

For a fixed positive $r \times c$ support, the plug-in estimator has a positive leading bias as the sample size grows:

$$\text{Bias}\{\widehat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \tag{2.8}$$

The leading term grows with table size and shrinks as the sample grows. It does not describe every small or sparse sample. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2} \left(\frac{1}{n_P} - \frac{1}{n_Q} \right). \tag{2.9}$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MIs. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

2.3 Sampling variance of plug-in MI

Comparing two MI estimates requires knowing how much each varies across repeated samples. We derive the large-sample approximation

$$\mathrm{Var}\{\widehat{I}(P)\} \approx \frac{V(P)}{n}. \tag{2.10}$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004).

The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_{i} p_{i+} \log p_{i+} - \sum_{j} p_{+j} \log p_{+j}. \tag{2.11}$$

Each sum contains the same function, $f(u) = u \log u$. Expanding this function lets us translate probability errors into changes in MI. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du}\{u \log u\} = \log u + u\frac{1}{u} = \log u + 1. \tag{2.12}$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \tag{2.13}$$

where a is the value around which the function is expanded and $x - a$ is the distance from that value. Here the new value is the estimated probability $x = \widehat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \widehat{u} - u$, which is the sampling error in the estimated probability. Substituting

these quantities into the general formula gives

$$\widehat{u}\log\widehat{u} \approx \underbrace{u\log u}_{\text{population value}} + \underbrace{(\log u + 1)(\widehat{u} - u)}_{\text{first-order change}}. \tag{2.14}$$

Subtracting the population value from both sides makes the change explicit:

$$\widehat{u}\log\widehat{u} - u\log u \approx (\log u + 1)(\widehat{u} - u). \tag{2.15}$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned}\widehat{I}(P) - I(P) \approx &\sum_{i,j}(\log p_{ij} + 1)(\widehat{p}_{ij} - p_{ij}) \\ &- \sum_{i}(\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) \\ &- \sum_{j}(\log p_{+j} + 1)(\widehat{p}_{+j} - p_{+j}).\end{aligned} \tag{2.16}$$

The margin errors are sums of the corresponding cell errors:

$$\widehat{p}_{i+} - p_{i+} = \sum_{j}(\widehat{p}_{ij} - p_{ij}), \qquad \widehat{p}_{+j} - p_{+j} = \sum_{i}(\widehat{p}_{ij} - p_{ij}). \tag{2.17}$$

For example, the row-margin sum can now be written as

$$\begin{aligned}\sum_{i}(\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) &= \sum_{i}\left[(\log p_{i+} + 1)\left\{\sum_{j}(\widehat{p}_{ij} - p_{ij})\right\}\right] \\ &= \sum_{i,j}(\log p_{i+} + 1)(\widehat{p}_{ij} - p_{ij}).\end{aligned} \tag{2.18}$$

The inner sum over j is the probability error for a fixed row i. It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way.

After this rearrangement, the same cell error $\widehat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned}&(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ &\quad = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right) - 1 \\ &\quad = \ell_P(i,j) - 1.\end{aligned} \tag{2.19}$$

Therefore,

$$\widehat{I}(P) - I(P) \approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}). \tag{2.20}$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j} (\widehat{p}_{ij} - p_{ij}) = \sum_{i,j} \widehat{p}_{ij} - \sum_{i,j} p_{ij} = 1 - 1 = 0. \tag{2.21}$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned} \widehat{I}(P) - I(P) &\approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\widehat{p}_{ij} - p_{ij}) - \sum_{i,j} (\widehat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.22}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a. An empirical cell probability is the proportion of observations in that cell, so

$$\widehat{p}_{ij} - p_{ij} = \frac{1}{n} \sum_{a=1}^{n} \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}. \tag{2.23}$$

Substituting this expression into Equation (2.22) first gives

$$\widehat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i,j) \left[\frac{1}{n} \sum_{a=1}^{n} \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\} \right]. \tag{2.24}$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned} \widehat{I}(P) - I(P) &\approx \frac{1}{n} \sum_{a=1}^{n} \left[\sum_{i,j} \ell_P(i,j) \mathbf{1}(Z_a = (i,j)) - \sum_{i,j} p_{ij} \ell_P(i,j) \right] \\ &= \frac{1}{n} \sum_{a=1}^{n} \{\ell_P(Z_a) - I(P)\}. \end{aligned} \tag{2.25}$$

The first sum inside the brackets selects the single cell containing Z_a, giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$.

To first order, Equation (2.25) expresses the MI error as an average of n observation-level contributions. The contribution of an observation in cell (i, j) is simply that cell's pointwise MI minus the population average, $\ell_P(i, j) - I(P)$. Its mean is zero because

$$\begin{aligned} \mathrm{E}_P\{\ell_P(X, Y) - I(P)\} &= \sum_{i,j} p_{ij}\{\ell_P(i, j) - I(P)\} \\ &= \underbrace{\sum_{i,j} p_{ij}\ell_P(i, j)}_{I(P)} - I(P) \underbrace{\sum_{i,j} p_{ij}}_{1} = 0. \end{aligned} \tag{2.26}$$

The centred function $\ell_P(i, j) - I(P)$ is called the influence function of MI. Equation (2.25) uses it to approximate the sampling error (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \mathrm{Var}_P\{\ell_P(X, Y) - I(P)\} = \sum_{i,j} p_{ij}\{\ell_P(i, j) - I(P)\}^2. \tag{2.27}$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned} \mathrm{Var}\{\widehat{I}(P)\} &\approx \mathrm{Var}\left\{\frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}\mathrm{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n} V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \tag{2.28}$$

Averaging n independent contributions therefore divides their variance by n. For dependent observations, covariance terms must also be included in Equation (2.28), as studied by Moddemeijer (1999). The present experiment uses independent observations.

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\widehat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0, 1). \tag{2.29}$$

This gives a large-sample normal approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i,j) = \log 1 = 0$ in every cell. Hence $I(P) = 0$, $V(P) = 0$, and the first-order average in Equation (2.25) vanishes. A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limiting distribution (Brillinger, 2004; Marinescu and Balcau, 2025). Testing independence therefore uses a chi-squared distribution rather than the normal approximation for comparing positive MI values.

2.4 Approaches to Testing MI Differences

This section reviews methods for estimating uncertainty in MI and testing differences between two MI values. The question in this thesis is whether two populations have the same MI, $I(P) = I(Q)$. Their cell probabilities may differ even when their MI values are equal.

Normal Wald inference.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with a standard normal distribution (Wald, 1943; van der Vaart, 1998). For two independent MI estimates, the two first-order variance contributions add:

$$\operatorname{Var}\{\widehat{I}(P) - \widehat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \tag{2.30}$$

Each term is the sampling variance of one MI estimate. Independence of the two samples makes their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\widehat{I}(P) - \widehat{I}(Q)}{\sqrt{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q}} \xrightarrow{d} N(0,1) \qquad \text{under } I(P) = I(Q). \tag{2.31}$$

Here $\widehat{V}(P)$ and $\widehat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. This limit assumes the regular positive-support conditions of Section 2.3 and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test. With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can

vary between samples. The Wald test still uses a normal distribution; it makes no separate small-sample adjustment for that standard-error variability.

The Welch–Satterthwaite principle.

Welch's test compares two means when their population variances may differ (Welch, 1947). For two independent samples with means $\bar{X}_P, \bar{X}_Q$ and variances s_P^2, s_Q^2, the statistic for testing equal population means and its effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \tag{2.32}$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\dfrac{(s_P^2/n_P)^2}{n_P - 1} + \dfrac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \tag{2.33}$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are $n_P - 1$ and $n_Q - 1$. If s_P^2/n_P dominates, the formula makes the combined ν close to $n_P - 1$; when both contributions matter, both component values affect ν. Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

Welch compares the statistic with a Student t distribution using the calculated ν. At the same significance level, this requires a larger absolute statistic to reject the null than a standard normal distribution does. The difference between the rejection thresholds from the Student t and standard normal distributions becomes smaller as ν increases. For MI, the estimated variance contributions are $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. Their component degrees of freedom need their own calculation because $\widehat{V}(P)$ and $\widehat{V}(Q)$ are nonlinear functions of the probability tables. Chapter 4 derives these MI-specific inputs to the same moment-matching structure.

This distinction matters because the size of a variance estimate and its stability answer different questions. The estimate $\widehat{V}(P)/n_P$ describes the expected spread of an MI estimate across samples. The variability of $\widehat{V}(P)$ describes how much that assessment of spread itself changes across samples. Two variance estimators with the same mean can have different variability. Satterthwaite's calculation uses both moments: greater variability relative to the squared mean gives fewer degrees of freedom and a larger rejection cutoff (Satterthwaite, 1946). A sample-size assignment alone does not adapt to differences in this relative variability between probability tables.

The same principle has been used in regression to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and

Carroll, 2001; Fay and Graubard, 2001; Imbens and Kolesár, 2016). For example, Pan and Wall (2002) calculate approximate degrees of freedom by examining how a regression variance estimate varies between samples. Expanded Welch applies this approach to the estimated MI variance. These regression precedents address variance estimates that are too small or confidence intervals that cover too rarely (Bell and McCaffrey, 2002; Kauermann and Carroll, 2001). A larger cutoff from the Student t distribution can widen the intervals and reduce excessive rejection in such settings. In the small-MI comparisons in Section 6.2, the average squared plug-in standard error instead exceeds the measured variance of the estimated difference. The larger cutoff leaves that variance error unchanged and makes an already-conservative test reject less often.

Prior MI inference and related comparisons.

Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. These calculations describe the uncertainty of an individual estimate, a necessary input to a comparison of two estimates. The first-order results of Moddemeijer (1989) and Brillinger (2004) give this uncertainty as $V(P)/n_P$, with $V(P)$ determined by the population table. Replacing its probabilities by observed proportions produces the variance estimate used by Wald. This supplies a standard error for the MI difference. Choosing a Welch–Satterthwaite cutoff requires one further quantity: how much the estimated variance varies when a new sample produces a different table.

A Bayesian approach describes uncertainty about population MI given an observed table. Hutter (2002) and Hutter and Zaffalon (2005) model the unknown cell probabilities to calculate the variance and shape of the resulting MI distribution. Our repeated-sampling comparison instead asks how often each test rejects when new samples are drawn from fixed populations.

Hutcheson's test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student t distribution, using

$$\nu_H = \frac{(s^2_{H,P} + s^2_{H,Q})^2}{(s^2_{H,P})^2/n_P + (s^2_{H,Q})^2/n_Q} \tag{2.34}$$

as its effective degrees of freedom (Hutcheson, 1970). Here each of $s^2_{H,P}$ and $s^2_{H,Q}$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Hutcheson's rule assigns degrees of freedom from sample sizes. Expanded Welch instead calculates how the estimated MI variance changes with the table, including the effects of a cell change on its row and column totals.

The entropy comparison establishes that a Student t distribution can be used to compare estimated information quantities, but MI requires more than substituting a different numerator. MI combines the row, column and joint entropies of the same observations. Their estimation errors are connected, because one cell count contributes to all three. The MI variance formulas in Section 2.3 account for that connection. The additional calculation in Expanded Welch follows it through to the estimated variance: a cell change alters both the weights and the pointwise-MI values whose weighted variance is being estimated. Thus Hutcheson supplies a close comparison framework, while the cell and margin derivatives supply the MI-specific inputs.

In a segregation application, Mora and Ruiz-Castillo (2009) develop inference for an MI index and discuss pairwise comparisons across groups and periods; the later peer-reviewed account places that MI index within the wider segregation literature (Mora and Ruiz-Castillo, 2011). By contrast, Guha and Chothia (2014) test whether two continuous distributions are identical using MI between pooled observations and their sample labels. For discrete data, Drake and Guha (2014) develop an MI-based k-sample test for equality of distributions. These tests compare complete distributions, not the MI of two variables within each population.

Resampling alternatives.

A bootstrap would draw many new samples from the observed tables to estimate how much the test statistic varies. This requires repeating the calculation many times (Efron, 1979). A simple permutation test instead combines the two samples and shuffles their group labels. That shuffle is justified when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only $I(P) = I(Q)$: the tables can differ while their MI values are equal. Shuffling labels could therefore test a stronger claim than this thesis asks about (Chung and Romano, 2013). Specialised permutation methods allow distinct populations in some parameter comparisons (Janssen, 1997; Chung and Romano, 2013); their validity for equal MI would need to be established.

2.5 Previous work and this thesis's contribution

Existing work provides the leading bias and sampling variance of MI (Miller, 1955; Moddemeijer, 1989; Brillinger, 2004; Paninski, 2003). Hutcheson's entropy comparison assigns degrees of freedom from sample sizes (Hutcheson, 1970), while Pan and Wall (2002) use the variability of a regression variance estimate to choose

degrees of freedom. Applying the latter approach to MI requires calculating how its estimated variance changes with the observed table.

These results address two successive sources of uncertainty. The MI sampling variance describes how much the estimated association changes between samples. The variability of the estimated MI variance describes how consistently those samples estimate that uncertainty. This thesis derives the second quantity and uses it within the established Welch–Satterthwaite framework. The resulting test retains the same MI difference and standard error as Normal Wald, allowing the experiment to isolate the consequences of its additional cutoff calculation.

A new observation changes the empirical cell probabilities, row and column totals, and pointwise MI values. The remaining calculation is how these changes affect the estimated MI variance. This thesis uses that calculation to choose the degrees of freedom for a Student t distribution, then evaluates whether it improves false positives and detection. That evaluation also needs to examine the shared estimates themselves. Paninski (2003) shows that finite-sample information estimates can retain substantial bias after a leading correction. A larger rejection cutoff does not remove such bias, nor does it replace a poor first-order variance approximation. The experiment therefore measures these errors alongside rejection, to distinguish a useful cutoff adjustment from a reduction in rejection that leaves the underlying estimation problem unchanged. Chapter 3 defines this shared statistic before Chapter 4 derives the additional variance calculation.

Chapter 3

The Two-Sample MI Test and Wald Baseline

Normal Wald and Expanded Welch share an estimated MI difference and standard error. This chapter defines their statistic and the Normal Wald p-value.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \qquad N^Q = (N_{ij}^Q) \tag{3.1}$$

be independent $r \times c$ count tables with totals n_P and n_Q. The row and column categories have the same definitions in both populations. The empirical probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \qquad \widehat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \tag{3.2}$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \tag{3.3}$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \tag{3.4}$$

The large-sample argument uses the following assumptions.

Assumption 1 (Sampling)**.** *Observations are independent and identically distributed within each population, and the two population samples are independent.*

Assumption 2 (Aligned fixed alphabets)**.** *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive cell probabilities)**.** *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. Such samples remain in the study; we record how often cells have zero observed counts to show how sparse the samples are.*

Assumption 4 (Positive MI)**.** *Both populations have positive MI: $I(P) > 0$ and $I(Q) > 0$. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive first-order variances $V(P) > 0$ and $V(Q) > 0$. It excludes exact independence, where the first-order approximation breaks down.*

Assumption 5 (Logarithm units)**.** *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\widehat{\ell}_P(i,j) = \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right) \qquad \text{when } \widehat{p}_{ij} > 0. \tag{3.5}$$

The plug-in MI estimate is

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\widehat{\ell}_P(i,j). \tag{3.6}$$

The same calculation gives $\widehat{I}(Q)$.

The leading fixed-support MI bias appears in Equation (2.8) (Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\widehat{\Delta}_{\mathrm{BC}} = \left\{\widehat{I}(P) - \frac{(r-1)(c-1)}{2n_P}\right\} - \left\{\widehat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q}\right\}. \tag{3.7}$$

The two terms in braces are the bias-corrected MI estimates for the separate populations. The correction uses the fixed numbers of row and column categories, even when some are unobserved in a sample. Corrected estimates can be negative; those values are retained when calculating the difference. When $n_P = n_Q$, the two corrections cancel in the difference; otherwise, their difference removes the leading unequal-sample bias described in Section 2.2.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\widehat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates $V(P)$ by

$$\widehat{V}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2. \tag{3.8}$$

Define $\widehat{V}(Q)$ analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}. \tag{3.9}$$

The common standardised statistic is

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}. \tag{3.10}$$

The sign of T indicates which sample has the larger corrected MI estimate. Its magnitude measures the difference relative to its estimated standard error. Under H_0, we compare T with a reference distribution centred at zero. Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

3.3 Normal Wald reference and motivation for expansion

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0,1) \qquad \text{under } H_0. \tag{3.11}$$

The two-sided Normal Wald p-value is

$$p_{\mathrm{Wald}} = 2\{1 - \Phi(|T|)\}, \tag{3.12}$$

where Φ is the standard normal distribution function.

Why consider an expanded reference?

Wald uses the standard normal distribution at every sample size, even though its standard error is estimated from the observed tables. Expanded Welch uses the variability of that variance estimate to choose a Student t distribution instead. Hutcheson's entropy test and ordinary Welch's test set each variance component's degrees of freedom from its sample size: n_i (Hutcheson, 1970) or $n_i - 1$ (Welch, 1947). Neither assignment measures how cell probabilities and margins make the estimated MI variance change between samples.

To choose the degrees of freedom, we need the average value of each estimated MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For P, to first order,

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P). \tag{3.13}$$

Chapter 4 derives $\mathrm{Var}\{\widehat{V}(P)\}$ by tracking how the calculated MI variance changes when one observation changes the joint table.

Chapter 4

The Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3. It changes the distribution used to judge that statistic from the standard normal distribution to a Student t distribution. To choose its degrees of freedom, this chapter calculates how much the estimated MI variance changes between samples.

4.1 Choosing Degrees of Freedom for Expanded Welch

The Welch–Satterthwaite equation combines the two estimated MI variance contributions (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\nu_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\nu_Q}. \tag{4.1}$$

The variance contributions $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$ already appear in the shared standard error. The new quantities are the component degrees of freedom ν_P and ν_Q. They describe how much each estimated MI variance varies between samples relative to its mean.

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of $\widehat{V}(P)$. Second, we calculate how $V(P)$ changes when a small amount of probability moves into one cell. Third, that change

tells us how variable $\widehat{V}(P)$ is. Finally, we combine the degrees of freedom from P and Q using Equation (4.1).

Moment matching for one component.

Satterthwaite approximates the positive variance estimate $\widehat{V}(P)$ by the scaled chi-squared variable

$$\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}. \tag{4.2}$$

The scale and degrees of freedom are chosen to match the mean and variance of $\widehat{V}(P)$.

For a chi-squared variable with ν_P degrees of freedom,

$$\mathrm{E}(\chi^2_{\nu_P}) = \nu_P, \qquad \mathrm{Var}(\chi^2_{\nu_P}) = 2\nu_P. \tag{4.3}$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$\mathrm{E}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}\right] = \mathrm{E}\{\widehat{V}(P)\}. \tag{4.4}$$

Its variance is

$$\mathrm{Var}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}\right] = \frac{[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_P^2}\,\mathrm{Var}(\chi^2_{\nu_P}) \tag{4.5}$$

$$= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_P}. \tag{4.6}$$

Matching this quantity to $\mathrm{Var}\{\widehat{V}(P)\}$ gives

$$\boxed{\nu_P = \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}}. \tag{4.7}$$

The numerator depends on the mean of the estimated MI variance; the denominator measures how much that estimate fluctuates between samples. To first order, $\mathrm{E}\{\widehat{V}(P)\} \approx V(P)$, so the remaining task is to calculate $\mathrm{Var}\{\widehat{V}(P)\}$.

4.2 How the MI Variance Changes with the Table

To find how much the estimated MI variance changes between samples, we first ask how the population variance responds when a small amount of probability is moved into one cell (Hampel, 1974; van der Vaart, 1998). Fix a cell (x, y) and define the perturbed distribution

$$P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}, \tag{4.8}$$

where $\delta_{(x,y)}$ places probability one on cell (x, y). At $\varepsilon = 0$, the distribution is P. Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell (i, j),

$$p_{ij}(\varepsilon) = (1-\varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \tag{4.9}$$

The joint and marginal probability derivatives at the original distribution are

$$\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \tag{4.10}$$

$$\left.\frac{\mathrm{d}p_{i+}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \tag{4.11}$$

$$\left.\frac{\mathrm{d}p_{+j}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \tag{4.12}$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j}\{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \tag{4.13}$$

Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \tag{4.14}$$

For each logarithm, the derivative is the probability derivative divided by the probability. Substituting the joint and marginal derivatives gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\ell_{P_\varepsilon}(i,j)\right|_0 &= \frac{\mathbf{1}\{i=x,j=y\}-p_{ij}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}-p_{i+}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}-p_{+j}}{p_{+j}} \\ &= \frac{\mathbf{1}\{i=x,j=y\}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}}{p_{+j}} + 1. \end{aligned} \tag{4.15}$$

The joint term supplies -1, while subtracting the row and column terms supplies $+1$ each. Their sum is the final $+1$.

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j). \tag{4.16}$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned}\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 &= \sum_{i,j}\{\mathbf{1}(i=x,j=y)-p_{ij}\}\ell_P(i,j) \\ &\quad + \sum_{i,j} p_{ij}\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 . \end{aligned} \tag{4.17}$$

The first sum is $\ell_P(x,y) - I(P)$. In the second sum, the selected-cell term contributes $p_{xy}/p_{xy} = 1$. The row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes -1; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \ell_P(x,y) - I(P)}. \tag{4.18}$$

This is the centred pointwise-MI influence function from Section 2.3.

Effect on the MI variance.

The target is the first-order change in

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y)\}. \tag{4.19}$$

Write this variance in second-moment form:

$$V(P) = \mathrm{E}_P\{\ell_P(X,Y)^2\} - I(P)^2. \tag{4.20}$$

Define the observation-level variance sensitivity by

$$g_P(x,y) = \left.\frac{\mathrm{d}V(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.21}$$

Differentiating Equation (4.20) gives

$$g_P(x,y) = \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 - 2I(P)\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.22}$$

It remains to differentiate

$$\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j)^2. \tag{4.23}$$

The product rule gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 &= \sum_{i,j}\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 \ell_P(i,j)^2 \\ &\quad + 2\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0.\end{aligned} \tag{4.24}$$

The first sum is

$$\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\}. \tag{4.25}$$

For the second sum, define the probability-weighted row and column means

$$\mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}}, \tag{4.26}$$

$$\mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}}. \tag{4.27}$$

Substitution of Equation (4.15) gives

$$\begin{aligned}&\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 \\ &\quad = \ell_P(x,y) - \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}} - \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}} + I(P) \\ &\quad = \ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} - \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P).\end{aligned} \tag{4.28}$$

Combining Equations (4.18), (4.22), (4.25), and (4.28) gives the variance sensitivity. To see how its squared term arises, combine the weight-change term in

Equation (4.25) with the derivative of $-I(P)^2$:

$$\begin{aligned}
&\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\} - 2I(P)\{\ell_P(x,y) - I(P)\} \\
&\quad = \ell_P(x,y)^2 - \{V(P) + I(P)^2\} - 2I(P)\ell_P(x,y) + 2I(P)^2 \\
&\quad = \ell_P(x,y)^2 - 2I(P)\ell_P(x,y) + I(P)^2 - V(P) \\
&\quad = \{\ell_P(x,y) - I(P)\}^2 - V(P).
\end{aligned}$$

The second line uses $\mathrm{E}_P\{\ell_P(X,Y)^2\} = V(P) + I(P)^2$; the last line completes the square. Adding twice the expression in Equation (4.28) gives

$$\boxed{\begin{aligned}
g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P) \\
+ 2\Big[\ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} \\
- \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)\Big].
\end{aligned}} \tag{4.29}$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for the fact that they change too: each pointwise-MI value uses a cell probability and its row and column totals. The conditional means collect the effects of those changing totals. Both parts are needed to describe how the estimated variance varies across samples.

4.3 Component degrees of freedom and final test

Population component degrees of freedom.

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}. \tag{4.30}$$

The sensitivities average to zero under P, as shown in Appendix A. Their variation, not their average, determines how much the estimated MI variance changes between samples.

Assumption 6 (Regular variance sensitivity)**.** *For the first-order Expanded Welch construction,* $\tau^2(P) > 0$ *and* $\tau^2(Q) > 0$.

An observation in cell $Z_a^{(P)}$ can be represented by a table that places probability one in that cell, denoted $\delta_{Z_a^{(P)}}$. Relative to P, that observation contributes the table change $\delta_{Z_a^{(P)}} - P$. The empirical table averages these one-observation tables, so its difference from P is

$$\widehat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \left\{\delta_{Z_a^{(P)}} - P\right\}. \tag{4.31}$$

The plug-in $\widehat{V}(P)$ applies the variance calculation to this empirical table. Equation (4.21) gives its first-order change for one observation's direction. Because a first-order change is linear in the table error (van der Vaart, 1998, Chapter 20), the changes from the observations can be averaged:

$$\widehat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \tag{4.32}$$

The observations are independent, so

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \mathrm{Var}_P\{g_P(Z_a^{(P)})\} \tag{4.33}$$

$$= \frac{\tau^2(P)}{n_P}. \tag{4.34}$$

Substitution into Equation (4.7) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \tag{4.35}$$

To interpret Equation (4.35), divide the sampling variance of $\widehat{V}(P)$ by the square of the population quantity it estimates:

$$\frac{\mathrm{Var}\{\widehat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \tag{4.36}$$

so the component degrees of freedom are approximately twice the reciprocal of this relative variance. When the estimate fluctuates little compared with the size of $V(P)$, the degrees of freedom are large and the Student t distribution is close to the standard normal distribution. Greater relative fluctuation gives fewer degrees of freedom and a larger rejection cutoff. Positive $V(P)$ does not guarantee positive $\tau^2(P)$: table changes can have no first-order effect on $V(P)$. In that case, describing the sampling variation of $\widehat{V}(P)$ requires a higher-order approximation.

Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.29) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \widehat{g}_P(i,j) = {} & \{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2 - \widehat{V}(P) \\ & + 2\left[\widehat{\ell}_P(i,j) - \frac{\sum_{j'} \widehat{p}_{ij'} \widehat{\ell}_P(i,j')}{\widehat{p}_{i+}} - \frac{\sum_{i'} \widehat{p}_{i'j} \widehat{\ell}_P(i',j)}{\widehat{p}_{+j}} + \widehat{I}(P)\right]. \end{aligned} \tag{4.37}$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical variance is centred explicitly:

$$\widehat{\tau}^2(P) = \sum_{i,j} \widehat{p}_{ij} \left\{ \widehat{g}_P(i,j) - \sum_{k,l} \widehat{p}_{kl} \widehat{g}_P(k,l) \right\}^2 . \tag{4.38}$$

The estimated component degree of freedom is

$$\boxed{\widehat{\nu}_P = \frac{2n_P \widehat{V}(P)^2}{\widehat{\tau}^2(P)}}. \tag{4.39}$$

Repeating the same calculation for the second table gives $\widehat{\nu}_Q$.

Combining the two components.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\boxed{\widehat{\nu}_{\text{expanded}} = \frac{\left\{ \widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q \right\}^2}{\left\{ \widehat{V}(P)/n_P \right\}^2 / \widehat{\nu}_P + \left\{ \widehat{V}(Q)/n_Q \right\}^2 / \widehat{\nu}_Q}}. \tag{4.40}$$

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2 \left\{ 1 - F_{t_{\widehat{\nu}_{\text{expanded}}}}(|T|) \right\}, \tag{4.41}$$

where F_{t_ν} denotes the cumulative distribution function of a Student t variable with ν degrees of freedom.

For a fixed finite T, the Student t distribution gives a two-sided p-value at least as large as the standard normal distribution used by Wald. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid. Thus every Expanded Welch rejection is also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. The correction cannot raise rejection where Wald is already too conservative. Section A.5 examines a zero-component case in the combined degrees-of-freedom formula.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \qquad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \tag{4.42}$$

Their totals are $n_P = 105$ and $n_Q = 100$. In the first table, cell $(1,1)$ has count 35, and its row and column each total 50. Its empirical pointwise MI is therefore

$$\widehat{\ell}_P(1,1) = \log\left\{\frac{35/105}{(50/105)(50/105)}\right\} = \log(1.47) \simeq 0.38526.$$

This cell contributes $(35/105)(0.38526)$ to the plug-in MI estimate. Summing the contributions from all nine cells gives $\widehat{I}(P) = 0.13693$; the same calculation for the second table gives $\widehat{I}(Q) = 0.25848$ nats. Since $(3-1)(3-1) = 4$, the corrected difference is

$$\widehat{\Delta}_{\mathrm{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \tag{4.43}$$

The variance calculation weights each squared deviation from the estimated MI by its cell probability. Cell $(1,1)$, for example, contributes $(35/105)(0.38526 - 0.13693)^2 \simeq 0.02056$ to $\widehat{V}(P)$. Summing over all cells gives $\widehat{V}(P) = 0.27189$ and $\widehat{V}(Q) = 0.43371$. Hence

$$\widehat{\mathrm{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \qquad T = -1.44898. \tag{4.44}$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the same cell illustrates the sensitivity calculation. The first row's pointwise-MI values are 0.38526, -0.51083 and -0.64436, weighted by its counts 35, 10 and 5. Its conditional mean is

$$\frac{35(0.38526) + 10(-0.51083) + 5(-0.64436)}{50} \simeq 0.10308.$$

The first column has the same conditional mean because this count table is symmetric. Substituting these values into Equation (4.37) gives

$$\begin{aligned}\widehat{g}_P(1,1) &\simeq (0.38526 - 0.13693)^2 - 0.27189 \\ &\quad + 2\{0.38526 - 0.10308 - 0.10308 + 0.13693\} \\ &\simeq 0.4218.\end{aligned}$$

The probability-weighted mean of the sensitivities is zero, so this cell contributes $(35/105)(0.4218)^2 \simeq 0.0593$ to $\widehat{\tau}^2(P)$. Repeating the calculation for all cells in both tables gives $\widehat{\tau}^2(P) = 0.93212$ and $\widehat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.39) gives component degrees of freedom

$$\widehat{\nu}_P = 16.65, \qquad \widehat{\nu}_Q = 45.86. \tag{4.45}$$

Combining them with Equation (4.40) gives $\widehat{\nu}_{\text{expanded}} = 59.03$ and $p = 0.15264$. Neither method rejects at the 5% level.

Operational algorithm and complexity.

Table 4.1 summarises one comparison.

Table 4.1: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\widehat{V}(P)$, $\widehat{V}(Q)$, the shared standard error, and T.
4	Compute the row and column pointwise-MI means needed in $\widehat{g}_P$ and $\widehat{g}_Q$.
5	Compute $\widehat{\tau}^2(P)$, $\widehat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.40).
7	Return the p-value from the Student t distribution, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$.

4.5 Behaviour and limitations

Near independence.

Consider a population table that approaches independence while its row and column margins stay fixed. At independence, MI is zero and its first-order change is also zero. The leading changes in MI, the pointwise-MI variance $V(P)$, and the variance sensitivity $\tau^2(P)$ can therefore be compared using a second-order expansion. Appendix A.2 gives the calculation and its fixed-margin assumptions. Close to independence on that path, it gives

$$V(P) \approx 2I(P), \qquad \tau^2(P) \approx 4V(P). \tag{4.46}$$

Substituting these local approximations into the component degrees-of-freedom formula gives

$$\nu_P = \frac{2n_P V(P)^2}{\tau^2(P)} \approx \frac{n_P V(P)}{2} \approx n_P I(P). \tag{4.47}$$

For a fixed sample size, this approximation makes the degrees of freedom smaller as population MI approaches zero. The cutoff from the Student t distribution then increases, so a larger absolute statistic is needed to reject equal MI. These approximations require the fixed-margin path, fixed table size and positive cell probabilities. Using them for samples also requires a useful first-order sampling approximation. At exact independence, quadratic sampling variation replaces the first-order term; Section A.4 derives that case.

Theoretical status and validity conditions.

Equations (4.18) and (4.29) are exact derivatives for positive population tables. They provide first-order approximations to the sampling errors in MI and its estimated variance. The scaled chi-squared model in Equation (4.2) and the Student t distribution for T are finite-sample approximations. The MI difference and standard error come from the same tables. A sample that changes the estimated difference can also change its standard error, and the degrees-of-freedom adjustment does not account for how the two move together. Appendix A.3 gives the first-order covariance between each population's MI and variance estimates. As sample sizes grow under regular conditions, the estimated degrees of freedom increase and the Student t distribution approaches the standard normal distribution used by Wald.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error. Expanded Welch also needs positive, finite component and combined degrees of freedom. A failed requirement gives an invalid calculation, not evidence for the null. Appendix B.4 specifies the numerical criteria; the experiment evaluates validity alongside calibration.

Chapter 5

Experimental Design

The experiment evaluates when Expanded Welch provides a useful adjustment to Normal Wald. It first measures false positives when the two populations have equal MI, then examines the estimation errors behind those results, and finally measures detection when MI differs. This sequence connects the test's behaviour to the quantities used in its calculation. Each comparison starts from fixed joint probability tables with known MI values, so the true difference is available when assessing the sampled estimates.

5.1 Questions and controlled comparisons

Three questions organise the study: how sample size and population structure affect false positives, how association strength affects the adjustment, and what the adjustment means for detection. Table 5.1 summarises the comparisons. Holding the other stated factors fixed makes each change easier to interpret. Specifying the questions, populations, methods and performance measures follows the simulation-design guidance of Morris et al. (2019).

The square table sizes are $2 \times 2, 4 \times 4, 8 \times 8$ and 16×16, containing 4, 16, 64 and 256 cells. Each doubling of the number of categories multiplies the number of cells by four. At a fixed sample size, observations are therefore distributed over more possible cells.

Three margin regimes compare uniform category frequencies, the same uneven frequencies in both populations, and different uneven frequencies. Under equal MI, the uniform and same-skew regimes provide identical-table controls. The different-skew regime tests comparisons between distinct population tables with equal association strength.

Table 5.1: The three controlled studies and supporting construction check.

Study	Factor varied	Main purpose
Equal MI	Sample size, table size and margins	False positives and usable results.
Association strength	Common positive MI	Degrees of freedom and near-independence behaviour.
Detection	True MI difference	Detection alongside null calibration.
Construction check	Copying or four-cell transfer	Sensitivity to a different association pattern.

5.2 Constructing joint distributions with known MI

The construction changes association while preserving the category frequencies. For one population, draw X with category probabilities $a = (a_1, \ldots, a_k)$. With probability λ, set $Y = X$; otherwise draw Y independently with the same probabilities a. Copying makes matching categories more likely than under independence. Both branches give Y the same marginal probabilities, so changing λ does not change the row or column totals.

For a diagonal cell, an observation can arrive through either branch. For an off-diagonal cell, it must arrive through the independent branch. Combining these probabilities gives

$$p_{ij}(\lambda) = (1 - \lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}, \qquad 0 \leq \lambda < 1, \tag{5.1}$$

where the indicator is one when the two categories match and zero otherwise. At $\lambda = 0$, the variables are independent and MI is zero. Increasing λ increases MI; the target MI determines the required copying probability. Appendix B gives the numerical search and shows why the margins remain fixed.

Constructing a 2×2 joint distribution with positive MI.

For P, take $a = (0.7, 0.3)$ and a copying probability of approximately 0.20347. Cell $(1, 1)$ requires $X = 1$, which has probability 0.7. Given $X = 1$, copying always gives $Y = 1$, whereas an independent draw gives $Y = 1$ with probability 0.7. Thus

$$p_{11} \simeq 0.7\{0.20347 + (1 - 0.20347)(0.7)\} = 0.53273,$$
$$p_{12} \simeq 0.7(1 - 0.20347)(0.3) = 0.16727.$$

The second equation has no copying term because copying cannot produce $X = 1, Y = 2$. The remaining cells follow the same rule. For Q, use $a = (0.8, 0.2)$ and a copying probability of approximately 0.26091. These choices give $I(P) = 0.02$ and $I(Q) = 0.03$ nats:

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \qquad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \tag{5.2}$$

Both joint distributions have positive MI, and their true difference has magnitude 0.01 nats. Sampling changes the estimated MI values, but the population difference used to assess detection remains 0.01 nats.

Table sizes and margin regimes.

Equation (5.1) applies to all four table sizes. For uneven margins, the first category has probability ρ, and the remaining mass is shared equally:

$$m_k(\rho) = \left(\rho, \frac{1 - \rho}{k - 1}, \ldots, \frac{1 - \rho}{k - 1}\right). \tag{5.3}$$

Each population uses the same vector for its row and column margins. Table 5.2 specifies the vectors for P and Q. The copying probabilities are chosen separately to achieve their respective MI targets. Equal MI therefore gives identical tables in the uniform and same-skew regimes, but different tables in the different-skew regime.

Table 5.2: Row and column margins for each population.

Regime	P: rows and columns	Q: rows and columns
Uniform	$(1/k, \ldots, 1/k)$	$(1/k, \ldots, 1/k)$
Same skew	$m_k(0.8)$	$m_k(0.8)$
Different skew	$m_k(0.7)$	$m_k(0.8)$

Increasing table size under the uneven-margin rule spreads the remaining mass over more categories. It changes both the number of cells and their probabilities. This comparison shows what happens under a common margin rule as the table grows. Uniform versus same skew instead compares even and uneven category frequencies at a fixed size; different skew allows the two populations' frequencies to differ as well.

5.3 The three studies and construction check

5.3.1 Sample size, table size and margins

The first study fixes $I(P) = I(Q) = 0.02$ nats and crosses all four sizes with all three margin regimes. For each of the twelve fixed population pairs, sample size varies over

$$n_P = n_Q \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}. \tag{5.4}$$

These 120 settings show whether the false-positive rate approaches 5% as more observations become available, and whether table size or margins change that progression. Valid-result rates distinguish a genuinely low false-positive rate from a failure to calculate the test.

The common target of 0.02 nats keeps association positive and comparable across these settings, while allowing all four sizes and three margin regimes to use the same target. It fixes the quantity being tested, not the difficulty of estimating it: tables with equal MI can have different cell probabilities and different MI sampling variances. Keeping each population pair fixed throughout the sample-size sweep separates improvement from more observations from a change in the population. The sweep includes small samples, intermediate samples and large-sample checks so that both calculation failures and convergence are visible.

5.3.2 Association strength and degrees of freedom

The second study keeps 4×4 tables and different-skew margins, and changes the common positive MI:

$$I(P) = I(Q) \in \{0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05\}. \tag{5.5}$$

Repeating this sweep at $n = 100$ and 1,000 separates the effect of association strength from a change in sample size. The estimated degrees of freedom connect

the observed rejection rates to the cutoff from the Student t distribution. All targets are positive, so this study approaches independence without evaluating its exact boundary.

Holding table size and margins fixed makes this a direct check of the method's association-strength dependence. The small positive targets examine where the first-order MI variance becomes small; the larger targets show whether the same tests behave differently away from that region. Two sample sizes show whether an association-strength pattern persists when more data are available.

5.3.3 Detection of a true MI difference

The third study fixes $I(P) = 0.02$ and sets $I(Q) = 0.02 + |\Delta|$, using

$$|\Delta| \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2\} \quad \text{nats.} \tag{5.6}$$

At $n = 1{,}000$ and 10,000, compare all four sizes under different skew, then all three margin regimes at 4×4. Keeping the zero-difference point in each curve makes detection interpretable alongside false positives. The signed thesis parameter is $\Delta = I(P) - I(Q)$, which is negative when Q has the larger MI; figures use its known nonnegative magnitude.

The two sample sizes connect detection to the null study at the same population settings. Keeping P's MI fixed makes each curve start from a common baseline as Q's MI increases. Small differences show the range in which the methods may disagree; the larger differences show whether detection eventually becomes high. Reading each curve from its zero-difference point prevents high rejection under alternatives from being mistaken for useful sensitivity when false positives are already excessive.

5.3.4 A supporting change in construction

A four-setting check asks whether representative findings persist when association is concentrated in four cells. Start from the independence table, add the same amount to cells $(1, 1)$ and $(2, 2)$, and subtract it from $(1, 2)$ and $(2, 1)$. Each affected row and column gains and loses the same amount. This transfer follows the sparse construction of Berrett and Samworth (2021, Section 3(b)).

Use 4×4 different-skew tables at $n = 1{,}000$ and 10,000, with MI pairs $(0.02, 0.02)$ and $(0.02, 0.03)$. Match the targets and margins to the corresponding copying populations. In a binary table the two constructions are equivalent; in a larger table

copying changes all diagonal and off-diagonal cells, whereas the transfer changes only four. This check tests that specific change in association pattern.

The complete design contains 68 population pairs and 232 distinct settings. Shared settings, such as the zero-difference points in the detection study, use the same simulation results in each comparison. At 20,000 repetitions per setting, the study samples 4,640,000 table pairs.

5.4 Sampling and performance measures

For each setting, draw 20,000 independent multinomial count-table pairs from its fixed P and Q. The samples from the two populations are independent, and both methods are applied to the same pair. Copying creates dependence between X and Y within an observation, not between the two population samples. The MI estimator, bias correction and standard error are those in Chapter 3; the Expanded Welch calculation is given in Chapter 4.

All settings use a two-sided 5% test. The primary rejection rate is

$$\frac{\#\{\text{valid p-values no larger than } 0.05\}}{20{,}000}. \tag{5.7}$$

At equal MI this estimates false positives; when MI differs it measures detection. Invalid calculations contribute no rejection but are not evidence for equal MI. The valid-result frequency is therefore reported alongside rejection. Rates restricted to method-valid or common-valid samples and their denominators are specified in Appendix B.

Wilson 95% intervals quantify simulation uncertainty for individual rates (Wilson, 1927). Intervals for the difference between methods use paired rejection indicators because the tests receive the same data. For 20,000 repetitions and a rate near 0.05, the Monte Carlo standard error is approximately 0.00154. Thus departures of only a few thousandths need to be read alongside the intervals. Section B.6 gives the calculations.

Diagnostics of the shared estimates.

The known populations also let us measure estimation error. Difference bias is the mean corrected estimate minus the true difference. The sample variance across repetitions measures its observed spread, and the mean squared standard error measures the uncertainty assigned by the tests. Their ratio indicates whether the estimated uncertainty is too large or too small on average. Comparing the observed

variance with $V(P)/n_P + V(Q)/n_Q$ separately checks the first-order approximation with its population inputs.

Degrees-of-freedom summaries use the median and 5th–95th percentile range among valid Expanded Welch outputs. This range describes differences between sampled results, not uncertainty in the estimated median. Empty-cell, row and column counts describe sparse sampling. The diagnostics retain finite estimates even if the associated test is invalid; Appendix B specifies the subsets used in each comparison. These measures help explain the curves without changing the test or excluding sparse settings.

Finally, a small timing comparison measures the computational cost on pre-generated 2×2 through 16×16 table pairs at $n = 1{,}000$ under different skew. Sampling and population construction are excluded from the timings. Both methods scan the cells, so their operation counts grow linearly with the number of cells. The timing procedure and measurement environment are reported with the results.

Chapter 6

Experimental Results and Discussion

The results follow the three experimental questions: when the tests give reliable false-positive rates, how association strength affects the adjustment, and how often a true MI difference is detected. Each rate uses 20,000 sampled table pairs. Blue solid curves show Normal Wald and magenta dashed curves show Expanded Welch. Error bars or bands give pointwise Wilson 95% intervals, and the dotted line marks 0.05. Hollow markers indicate valid-result frequency below 90%. The parameter settings are those in Chapter 5; additional comparisons and the full 120-setting null table are in Appendix C.

6.1 False positives across sample size and population structure

6.1.1 Increasing table size

Figure 6.1 holds both population MI values at 0.02 nats and compares the four table sizes under different skew. The dominant marginal probabilities are 0.7 for P and 0.8 for Q, with the remaining mass shared equally across their other categories. Each panel varies only sample size, from 10 to 50,000 per population. The lower strip shows how often each method returns a valid p-value.

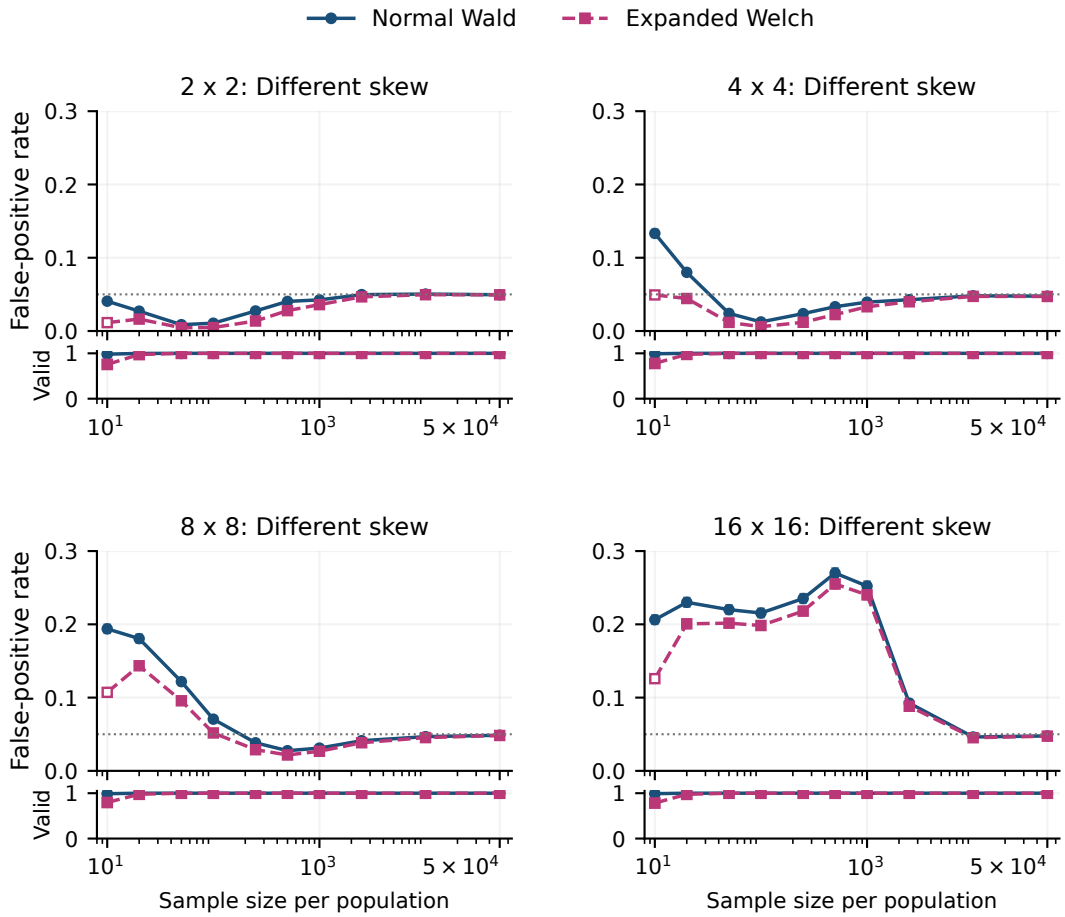


Figure 6.1: False positives and validity under different-skew margins.

The false-positive pattern changes substantially with table size. In 2×2 tables both methods reject too rarely over much of the sample-size range. In 4×4 tables Wald is above 0.05 at $n = 10$ and 20, but both methods fall below the target at larger intermediate sample sizes. The 8×8 comparison makes this transition later. Expanded Welch lowers rejection throughout, helping where Wald is excessive and moving an already-low rate further below 0.05.

At the smallest sample size, part of that reduction involves lost validity. For 4×4 tables at $n = 10$, Wald rejects at 0.1332 and Expanded Welch at 0.0493, but their valid-result frequencies are 0.9910 and 0.7776. Restricting both tests to the 15,552 common-valid pairs gives rejection rates 0.1053 and 0.0633. The Student t distribution still reduces rejection on those same pairs, but the unconditional rate near 0.05 also reflects failed calculations. Validity and calibration therefore answer different questions.

The 16×16 table shows that valid calculations need not be well calibrated. At $n = 1{,}000$, every calculation is valid, yet Wald rejects at 0.2522 and Expanded Welch at 0.2402. Their Wilson intervals are approximately [0.2462, 0.2583] and

[0.2343, 0.2462]. The paired reduction is 0.0120, with approximate 95% interval [0.0105, 0.0135], so the adjustment lowers rejection but leaves a large excess of false positives. At $n = 10{,}000$ the rates are 0.0463 and 0.0453; at 50,000 they are 0.0477 and 0.0474. More observations substantially improve this setting, although the path is not a simple steady decline. The next comparison asks how much of this pattern changes with the margins.

6.1.2 Changing the margin regime

Figure 6.2 compares all three margin regimes at 4×4 and 16×16, retaining the common MI target and sample-size sweep. Uniform and same-skew populations have identical joint distributions within each pair. Different-skew populations have different tables but equal MI. Comparing the columns shows how uneven category frequencies and different population margins change the test.

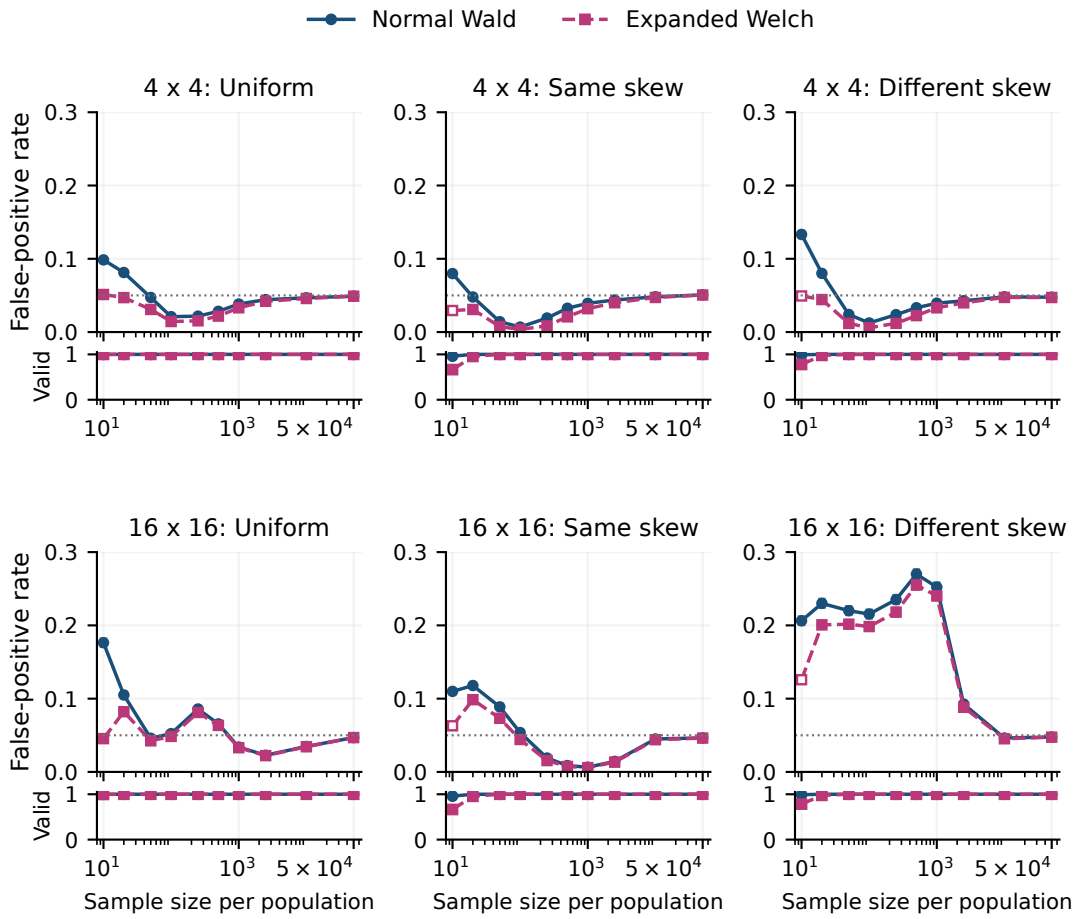


Figure 6.2: False positives and validity across three margin regimes.

At 4×4, all three regimes give low rejection at intermediate sample sizes and rates much nearer 0.05 by $n = 10{,}000$. At $n = 1{,}000$, for example, Wald rates

range from 0.0383 to 0.0394, while Expanded Welch ranges from 0.0319 to 0.0331. Changing the margins has a relatively small effect on these particular rates, and both methods are fully valid. The larger table behaves differently: at $n = 1{,}000$, Wald rejects at 0.0339 under uniform margins, 0.0065 under same skew, and 0.2522 under different skew. Expanded Welch gives 0.0331, 0.0059 and 0.2402, respectively. The different-skew excess is therefore not a general consequence of uneven margins alone.

The shared estimates help explain this contrast. Table 6.1 reports the four preselected different-skew null settings. Observed SD is the sample standard deviation of the corrected MI difference. Estimated SD is the square root of the mean squared standard error, rather than the mean standard error. The variance ratio divides the latter variance estimate by the observed variance. All four settings have full validity and finite diagnostics.

Table 6.1: Bias and uncertainty of the estimated MI difference.

Size	n	Bias	Observed SD	Estimated SD	Variance ratio
4×4	1,000	-0.00012	0.01053	0.01085	1.061
4×4	10,000	-0.00001	0.00316	0.00318	1.014
16×16	1,000	0.02960	0.01449	0.02019	1.942
16×16	10,000	-0.00088	0.00387	0.00401	1.073

The 16×16, $n = 1{,}000$ difference has bias 0.02960 nats, more than twice its observed SD of approximately 0.01448. To see where this shift comes from, recall that bias is the mean estimate across samples minus the population value. The bias of a difference is exactly the first estimate's bias minus the second estimate's bias. Here the individual corrected MI biases are approximately -0.00794 for P and -0.03754 for Q. Since the true difference is zero, the estimated difference is centred at

$$\mathbb{E}(\widehat{\Delta}_{\mathrm{BC}}) \simeq 0 + (-0.00794) - (-0.03754) = 0.02960 \quad \text{nats.} \tag{6.1}$$

Both corrected estimates are too low on average, but Q's is lower by more. Subtracting it therefore creates a positive difference even though the population MI values match. The common leading correction cancels because the table and sample sizes match; it cannot remove the remaining difference in biases. This is the finite-sample problem highlighted by Paninski (2003): a leading bias correction need not remove the error in sparsely sampled information estimates.

On average, 147 of P's 256 cells and 183 of Q's cells are empty. The shifted difference and sparse sampling occur together, while the observed individual biases directly identify the source of the shift.

Estimated uncertainty is also inaccurate in this setting: mean squared standard error is about 1.94 times the observed variance. Excessive false positives nevertheless remain, because a large standard error does not restore a difference estimate centred away from zero. At $n = 10{,}000$ the bias is much smaller and the variance ratio is 1.07, accompanying the improved null rate. The population first-order variance is about 0.60 times the observed variance at $n = 1{,}000$, compared with 0.84 at 10,000, showing that the approximation itself also needs attention. These diagnostics describe distinct errors; they do not divide the rejection rate into additive contributions. Expanded Welch changes none of these shared estimates. Its separate role is to change the rejection cutoff, which the association-strength study examines next.

6.2 Association strength and degrees of freedom

Figure 6.3 keeps 4×4 different-skew tables and varies the common population MI from 0.0001 to 0.05 nats. The two columns use $n = 100$ and 1,000. Upper panels show false positives; lower panels show the median estimated combined degrees of freedom, with the 5th–95th percentile range shaded. Every calculation in this sweep is valid, so the low rejection rates do not reflect missing test outputs.

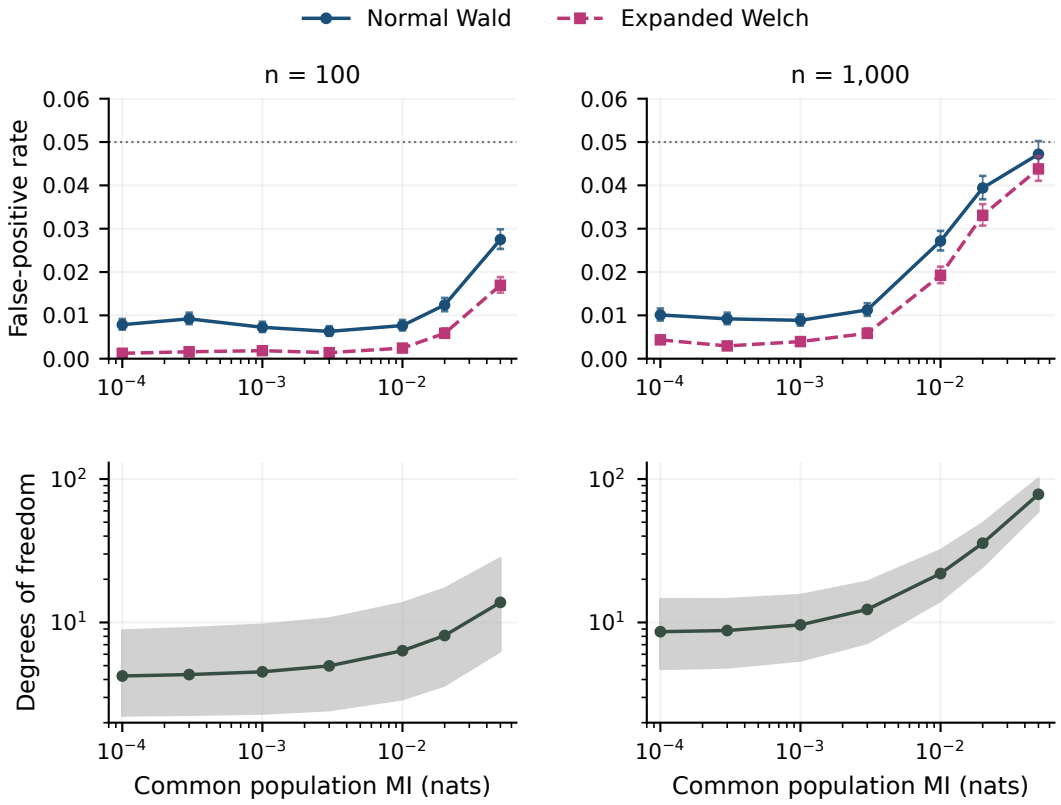


Figure 6.3: Association strength, false positives and estimated degrees of freedom.

Both tests reject too rarely near independence, and Expanded Welch rejects still less. At common MI 0.0001 and $n = 1{,}000$, Wald's rate is 0.0101 and Expanded Welch's is 0.0044. At common MI 0.05, the rates are 0.0472 and 0.0438. Stronger association brings the rates closer to 0.05 in this comparison, especially at the larger sample size. The $n = 100$ rates remain low even at the largest MI target, so a larger MI value does not substitute for enough observations.

Estimated degrees of freedom increase with association strength. At $n = 1{,}000$, their median rises from 8.62 at MI 0.0001 to 78.24 at MI 0.05; at $n = 100$, it rises from 4.23 to 13.80. Smaller degrees of freedom give a larger cutoff from the Student t distribution, requiring a larger absolute statistic for rejection. The near-independence calculation in Section 4.5 explains why small positive MI can make this adjustment restrictive. Here it acts on a Wald test that is already conservative.

The median degrees of freedom illustrate the change in cutoff. For a two-sided 5% test, the cutoffs from the Student t distribution are approximately 2.277 at 8.62 degrees of freedom and 1.991 at 78.24, compared with Normal Wald's 1.960. Thus $|T| = 2.1$ would reject under Wald but not under Expanded Welch with 8.62 degrees of freedom; at 78.24 both would reject. These examples use the reported medians. Each sampled table pair uses its own estimated degrees of freedom.

The estimation diagnostics show that the reference distribution is not the only issue. At MI 0.0001 and $n = 1{,}000$, mean squared standard error is 1.74 times the observed variance. The population first-order variance is only about 0.04 times that variance, indicating that the first-order approximation is poor this close to independence at this sample size. These ratios compare two different calculations. The population calculation uses the true probabilities in $V(P)/n_P + V(Q)/n_Q$. The estimated calculation replaces them by each sample's observed proportions before averaging the resulting squared standard errors. They need not have the same error.

Near independence, the true first-order variance is small because the population cell probabilities are close to the products of their margins. A finite sample can still show apparent association: observed cell proportions generally differ from the products of their observed margins. The plug-in calculation treats those departures as features of an estimated population table and can therefore assign a substantially larger first-order variance. Meanwhile, the linear approximation at the true table omits the quadratic sampling terms that matter near independence. Brillinger (2004) distinguishes these first-order and independence calculations; Section A.4 shows why the quadratic terms remain when the first-order term vanishes. Thus a population first-order variance that is too small can coexist with a plug-in variance estimate that is too large. The observed ratios show both errors in this setting.

At MI 0.05 the estimated/observed ratio is approximately 1.00 and the population/observed ratio is 0.95. Detailed values are in Appendix C. Improving the cutoff alone cannot correct errors in the statistic being compared with it. We now examine what that means when the populations have different MI values.

6.3 Detection when population MI differs

6.3.1 Detection across table sizes

Figure 6.4 fixes $I(P) = 0.02$ nats, sets $I(Q) = 0.02 + |\Delta|$, and uses different-skew margins. Each panel compares $n = 1{,}000$ and 10,000 at one table size. The figure enlarges differences up to 0.02 nats; the full grid through 0.2 is in Figure C.2. Every displayed detection setting is fully valid.

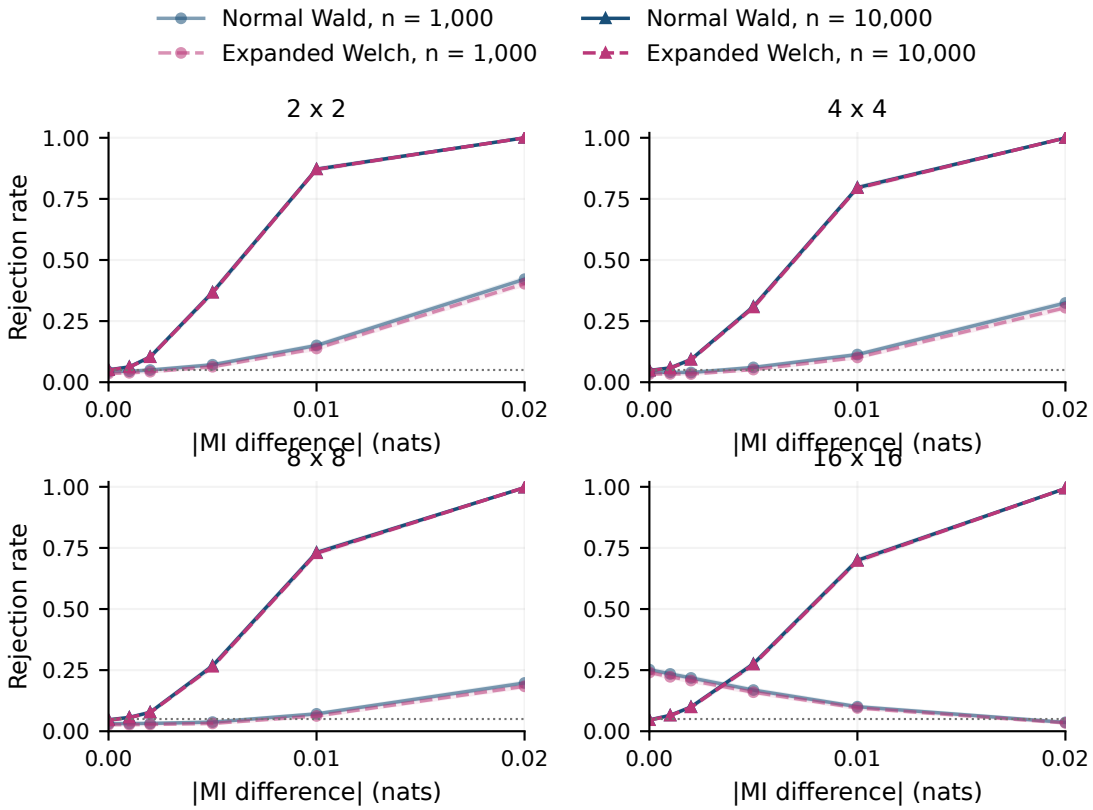


Figure 6.4: Detection over small MI differences under different-skew margins.

For the smaller tables, increasing the true difference generally increases detection, and larger samples make a clear improvement. In 4×4 tables, a 0.01-nat difference is detected at rates 0.1132 and 0.1013 by Wald and Expanded Welch at $n = 1{,}000$, compared with 0.7962 and 0.7940 at $n = 10{,}000$. The adjustment costs some detection at the smaller sample size and very little at the larger one. This follows its larger degrees of freedom in larger samples. The null points remain essential: these curves compare tests at the same nominal threshold, not at equal achieved false-positive rates.

The 16×16, $n = 1{,}000$ curve instead falls initially as the true difference grows. Wald rejects at 0.2522 under equal MI, 0.1005 at a difference of 0.01, and 0.0363 at 0.02. This is not improved calibration under those alternatives. The positive bias identified earlier opposes the negative true difference. The mean estimated difference is approximately 0.02960 at the null. Under the next two alternatives, the true difference and estimated difference bias combine as follows:

$$\mathbb{E}(\widehat{\Delta}_{\mathrm{BC}}) \simeq -0.01 + 0.03061 = 0.02061, \tag{6.2}$$

$$\mathbb{E}(\widehat{\Delta}_{\mathrm{BC}}) \simeq -0.02 + 0.03164 = 0.01164. \tag{6.3}$$

The bias changes slightly between these populations, but remains positive and larger than the magnitude of the true difference. As the true difference becomes more negative, it offsets more of that bias. The centre of the estimated difference therefore moves towards zero rather than away from it. Dividing by the estimated standard error can then give smaller absolute statistics, despite the larger population difference. The declining rejection rate is a consequence of comparing biased estimates, not evidence that the two population MI values have become more similar. The full-range comparison shows rejection rising again at larger differences.

At $n = 10{,}000$, the 16×16 bias is much smaller, and detection increases across the small differences. Wald detects a 0.01-nat difference at 0.6992 and Expanded Welch at 0.6968; by 0.02 the rates are 0.9932 and 0.9931. The larger sample therefore improves both the null behaviour and the detection pattern. High rejection in the smaller sample was not evidence of a more sensitive test.

6.3.2 Detection across margin regimes

Figure 6.5 fixes the table size at 4×4, retains the same MI targets, and compares the three margin regimes at the two sample sizes. This separates their effect on detection from the table-size comparison.

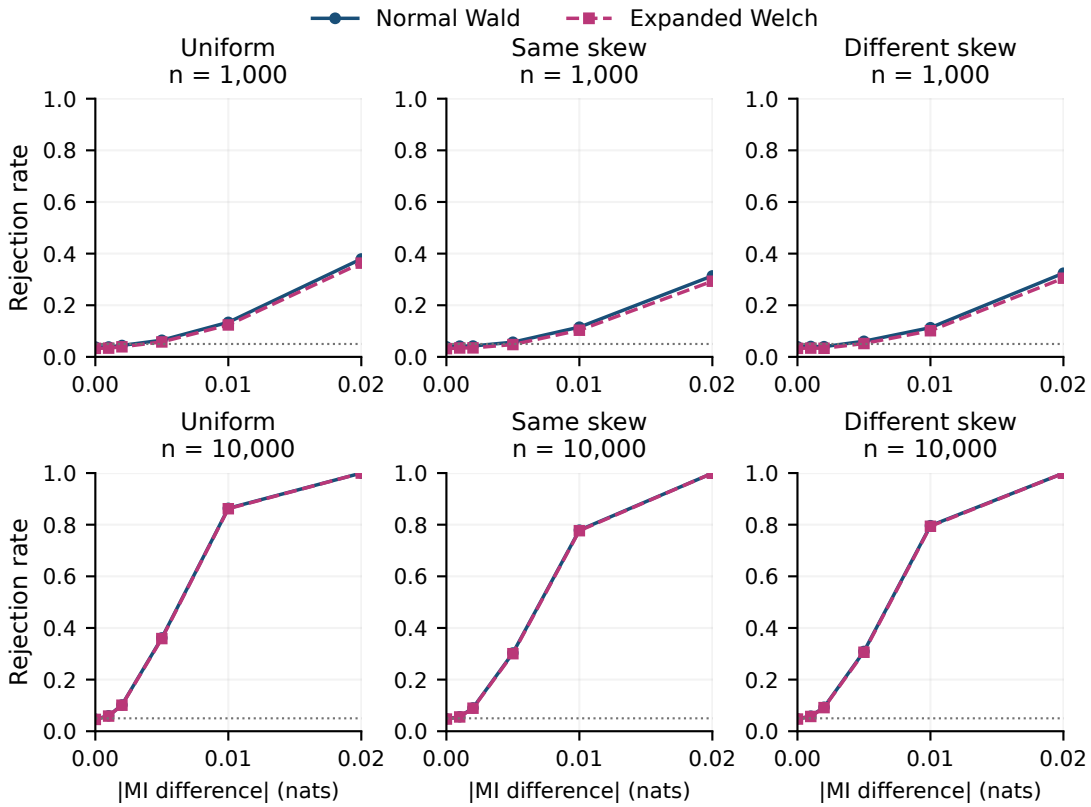


Figure 6.5: Detection over small MI differences in 4×4 tables.

Uniform margins give somewhat higher detection in these settings. At $n = 10{,}000$ and a 0.01-nat difference, Wald detects at 0.8635 under uniform margins, 0.7790 under same skew, and 0.7962 under different skew. Expanded Welch gives 0.8619, 0.7768 and 0.7940. The margin-related change is larger than the difference between tests at this sample size.

The observed precision helps explain the detection ordering. For this 0.01-nat difference at $n = 10{,}000$, the sample SD of the estimated difference is approximately 0.00327 nats under uniform margins, 0.00364 under same skew and 0.00361 under different skew. The biases are small in all three settings, and the square roots of the mean squared standard errors are close to these observed SDs. The same population difference is therefore easier to distinguish from sampling variation under the uniform construction. Equal MI targets do not imply equal precision: the cell probabilities also determine the variance in Equation (2.30). This connects the margin comparison to the first-order MI variance, rather than to the choice of rejection cutoff alone.

At $n = 1{,}000$, the adjustment produces a more visible detection reduction, while all three null points are already below 0.05. By a difference of 0.2 nats, both methods reject every simulated pair in all three regimes at both sample sizes.

The supporting four-cell check gives the same qualitative comparison at its matched 4×4 settings: the adjustment lowers an already-low null rate at $n =$ 1,000, and changes detection little at 10,000. Appendix C gives the rates and diagnostics. This agreement supports the representative comparison across those two constructions, while the main curves retain their specific copying populations.

6.4 Computation and practical implications

Table 6.2 reports scalar-call timings on an arm64 machine running macOS 26.6.2, Python 3.13.5, NumPy 2.4.4 and SciPy 1.17.1. Each setting uses 200 pre-generated different-skew null table pairs at $n = 1{,}000$, 20 warm-up calls per method and five timed passes. Calls are timed individually, the method order alternates between passes, and the table shows the median and interquartile range of 1,000 calls.

Table 6.2: Time per test call in microseconds: median [interquartile range].

Size	Normal Wald	Expanded Welch
2×2	84.23 [81.67, 85.42]	146.92 [143.12, 149.92]
4×4	84.67 [80.83, 86.01]	148.46 [144.67, 150.75]
8×8	86.04 [84.17, 87.17]	151.25 [146.78, 153.80]
16×16	91.67 [88.10, 92.87]	158.98 [154.53, 162.00]

Expanded Welch costs approximately 1.7–1.8 times as much in these timings. Both methods require $O(k^2)$ work for a $k \times k$ table, but fixed call overhead can be important for these small arrays. The statistical trade-off is more consequential than this modest absolute cost: the adjustment helps only when its lower rejection improves an excessive false-positive rate without too much loss of detection or validity. Larger samples reduce several shared estimation errors, whereas changing the reference distribution alone leaves those errors in place.

Chapter 7

Implications and Limitations

7.1 Implications for using Expanded Welch

Normal Wald and Expanded Welch use the same estimated MI difference and standard error. Expanded Welch can only make a valid two-sided p-value larger, reducing rejection for the same sampled tables. When Wald's false-positive rate exceeds 5%, reducing it towards 5% can help. When Wald already rejects too rarely, the same adjustment moves the rate further from the target and can reduce detection. A comparison of the two methods therefore needs both measures: false positives show whether the reduction improves calibration, and detection shows its cost when population MI differs.

The controlled studies show why a general rejection reduction is not a general improvement. At small positive MI, Wald is already conservative in the association-strength sweep, and Expanded Welch reduces rejection further. At 16×16 with different skew and $n = 1{,}000$, both tests instead have excessive false positives despite full validity. There the adjustment provides only a small reduction. By $n = 10{,}000$, the bias and variance diagnostics improve and the methods give similar detection.

Unequal estimation biases can produce an apparent change in association when population MI is equal. In Equation (6.1), the larger different-skew tables produce an estimated difference centred above zero even under equal MI. Increasing the rejection cutoff leaves that centre unchanged. Reliable decisions in such settings require attention to the MI estimates, not only to which distribution supplies the cutoff.

A usable calculation and a well-calibrated test are also different requirements. Validity here means that the quantities needed for the calculation are finite and satisfy the method's positivity conditions. It does not mean that the resulting p-values achieve the intended false-positive rate. The fully valid 16×16 comparison

at $n = 1{,}000$ demonstrates that distinction. Conversely, sparse samples can prevent Expanded Welch from returning a p-value, lowering its reported rejection rate without providing a decision about equal MI. Rejection and valid-result frequency therefore need to be read together.

For a categorical feature–label comparison, the estimated MI difference reports the size and direction of the observed change. Its standard error describes the sampling uncertainty assigned by the method, and the p-value supplies the test decision. Reporting these quantities together makes the comparison more informative than reporting a p-value alone. If Expanded Welch is used, its estimated degrees of freedom explain the additional cutoff adjustment. A small p-value is evidence against equal MI within the test's approximation; a large p-value does not establish that the population MI values are equal.

Normal Wald remains the simpler baseline, but its reliability also depends on the population setting. Expanded Welch can provide a supplementary comparison where simulations representative of the application show excessive Wald rejection and frequent usable Expanded Welch results. Such an assessment should match the category definitions, sample sizes and margin patterns, and vary plausible association strengths. Equal-MI pairs with different tables are important when the application's populations may differ. The assessment should compare false positives, detection and usable results, rather than selecting the method with fewer rejections alone. The present study supplies controlled examples of this reasoning, not a universal sample-size threshold. Method choice should be set before examining the observed p-values.

The main obstacle is statistical rather than computational. An Expanded Welch call takes approximately 1.7–1.8 times as long as a Wald call in the measured implementation, with medians below 160 microseconds over the tested sizes. Both have $O(rc)$ operation growth for an $r \times c$ table. The extra calculation is modest, but it does not remove bias or inaccurate uncertainty in the shared statistic.

7.2 Limits of the evidence

The rejection rates and Monte Carlo intervals apply to the constructed population pairs. The intervals quantify uncertainty from the finite number of simulated samples; they do not describe how performance varies across untested populations.

The primary populations use a copying mixture with identical row and column margins within each population. This gives a clear way to vary association without changing category frequencies. The four-cell check supports the direction of the

method comparison at its four matched settings, while showing that detection can still change numerically.

Copying favours matching categories in both populations, whereas the four-cell transfer concentrates association in the same chosen block. The different-skew regime changes category frequencies while keeping those association locations aligned. Its results therefore demonstrate that distinct equal-MI tables can be difficult to compare, but do not describe every way two tables with equal MI can differ. A broader assessment would vary association locations as well as their strengths and margins, including unequal row and column margins within a population.

Equal sample sizes simplify the comparison by making the two leading bias corrections cancel in $\widehat{\Delta}_{\mathrm{BC}}$. The remaining difference bias in the experiment consequently comes from estimation errors not removed by that common correction. With unequal sample sizes, the corrections no longer cancel, and the two contributions to the standard error and degrees of freedom can become less balanced. The derivation allows unequal sample sizes, but the experiment does not assess how well those additional adjustments perform. That requires a separate sample allocation study.

The sampling assumptions also limit transfer to other applications. Paired observations would require a covariance term in the variance of the difference, whereas the present samples are independent. Continuous MI estimators have different estimation errors from fixed categorical tables. Similarly, increasing the number of categories with sample size changes the fixed-alphabet setting of the derivation. The empirical recommendation concerns the independent categorical comparisons studied here, at a two-sided 5% threshold; a real-data evaluation would test how well these constructed comparisons represent an application.

Exact independence needs a different mathematical approximation because the first-order MI variance vanishes there. The association-strength study also shows that being positive is not enough to make that approximation accurate at a particular sample size: small positive MI can remain close to the problematic region. This explains why improving the first-order cutoff and improving the approximation near independence are separate tasks. Section A.4 derives the second-order approximation needed there.

Chapter 8

Conclusion

This thesis derives an Expanded Welch–Satterthwaite test for comparing MI between independent categorical populations. Its mathematical contribution is the observation-level sensitivity of the estimated MI variance, which determines the degrees of freedom. Its empirical contribution is a controlled 232-setting comparison with Normal Wald. Three connected studies, estimation diagnostics and a supporting construction check explain the observed false positives and detection.

Together, these contributions connect the stability of an estimated MI variance to an explicit, computable cutoff adjustment. They also show why that adjustment must be evaluated alongside the accuracy of the MI difference and standard error. This separates three questions that can otherwise be confused: whether the effect estimate is centred correctly, whether its uncertainty is estimated accurately, and whether the reference distribution gives a useful rejection threshold.

Expanded Welch lowers rejection, but its benefit depends on the errors already present in Wald. Near independence it makes conservative rates lower still. In larger different-skew tables, unequal MI bias can instead produce excessive false positives and an initial decline in detection as the true difference grows. The adjustment leaves the shared estimates unchanged. Larger samples improve the estimation diagnostics and observed performance, and the tests become similar. Normal Wald remains the simpler baseline; Expanded Welch is a supplementary check where a closely matched calibration study supports its use.

The evaluation covers four square table sizes, three margin regimes and equal sample sizes, primarily under a copying construction. The derivation requires positive cell probabilities and positive first-order MI variance; exact independence needs a different approximation. These conditions bound the practical recommendation.

Future work should first address bias in sparse MI differences and the dependence between the difference and its estimated standard error. The falling detection curve gives a concrete target: in these sparse settings, an improved comparison should avoid moving the estimated difference towards zero when the population difference grows. Alternative bias treatments should therefore be assessed through both null calibration and the shape of detection curves, rather than through MI estimation error alone.

Near independence, a second-order comparison may be more appropriate than refining the first-order reference distribution. A likelihood-ratio or resampling method would need validation for equal MI even when the complete population tables differ. This would address the approximation problem revealed by the association-strength diagnostics. Unequal sample sizes, other association patterns and a real-data application would then test the practical reach of these conclusions.

Appendix A

Supporting Calculations

These calculations support the variance sensitivity, near-independence interpretation and validity conditions in Chapter 4.

A.1 Mean of the variance influence function

The observation-level sensitivities $g_P(x, y)$ have a probability-weighted mean of zero. Averaging the one-cell perturbation directions leaves the population table unchanged:

$$\sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\} = P - P = 0. \tag{A.1}$$

This weighted cancellation also holds after exact plug-in substitution. The first derivative is linear in the direction of change. Therefore, the weighted average of the one-cell derivatives is the derivative in the weighted-average direction:

$$\begin{aligned} \mathrm{E}_P\{g_P(X, Y)\} &= \sum_{x,y} p_{xy} \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V\{(1-\varepsilon)P + \varepsilon\delta_{(x,y)}\}\Big|_0 && \text{(A.2)} \\ &= \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V\left\{P + \varepsilon \sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\}\right\}\Big|_0 && \text{(A.3)} \\ &= \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V(P)\Big|_0 = 0. && \text{(A.4)} \end{aligned}$$

A.2 Fixed-margin behaviour near independence

Let $P_0 = (q_{ij})$ be independent, with positive cells $q_{ij} = a_i b_j$. Consider $P_t = P_0 + tH$, where every row and column sum of H is zero. For sufficiently small t, all cells remain positive and the margins remain a and b. We expand around $t = 0$, where $I(P_0) = 0$.

For $f(t) = I(P_t)$, the second-order Taylor formula is

$$f(t) = f(0) + f'(0)t + \frac{f''(0)}{2!}t^2 + O(t^3). \tag{A.5}$$

Here $O(t^3)$ collects the omitted terms whose magnitude is bounded by a constant times $|t|^3$ as t approaches zero. These terms become smaller than the retained quadratic term. The same notation below describes the size of each omitted remainder near zero.

The fixed margins make the pointwise MI in cell (i, j)

$$\ell_{P_t}(i,j) = \log\left(\frac{q_{ij} + tH_{ij}}{q_{ij}}\right) = \log\left(1 + \frac{tH_{ij}}{q_{ij}}\right) = \frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3), \tag{A.6}$$

using $\log(1 + z) = z - z^2/2 + O(z^3)$ with $z = tH_{ij}/q_{ij}$. Multiplying by the cell probability gives

$$\begin{aligned}(q_{ij} + tH_{ij})\ell_{P_t}(i,j) &= (q_{ij} + tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3)\right\} \\ &= tH_{ij} - \frac{t^2H_{ij}^2}{2q_{ij}} + \frac{t^2H_{ij}^2}{q_{ij}} + O(t^3) \\ &= tH_{ij} + \frac{t^2H_{ij}^2}{2q_{ij}} + O(t^3).\end{aligned}$$

Multiplication by q_{ij} supplies the negative quadratic term; multiplication by tH_{ij} supplies the positive one. Their combination leaves half the positive term. When we sum over cells, the first-order terms cancel because $\sum_{i,j} H_{ij} = 0$. Thus $f(0) = f'(0) = 0$, $f''(0) = \sum_{i,j} H_{ij}^2/q_{ij}$, and

$$I(P_t) = \frac{t^2}{2}\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.7}$$

To calculate $V(P_t)$, first square the pointwise-MI expansion:

$$\ell_{P_t}(i,j)^2 = \left\{\frac{tH_{ij}}{q_{ij}} + O(t^2)\right\}^2 = \frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3).$$

Multiplying by the cell probabilities and summing gives its second moment:

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y)^2\} &= \sum_{i,j}(q_{ij} + tH_{ij})\left\{\frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3)\right\} \\ &= t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned}$$

MI itself begins with a quadratic term, so $I(P_t)^2 = O(t^4)$. Subtracting it from this second moment leaves the quadratic term unchanged:

$$V(P_t) = t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.8}$$

The row conditional mean also has no first-order term. Its denominator remains a_i because the margins are fixed, and

$$\begin{aligned} \mathrm{E}_{P_t}\{\ell_{P_t}(X,Y) \mid X = i\} &= \frac{1}{a_i} \sum_j (q_{ij} + tH_{ij}) \left\{ \frac{tH_{ij}}{q_{ij}} + O(t^2) \right\} \\ &= \frac{t}{a_i} \sum_j H_{ij} + O(t^2) = O(t^2). \end{aligned}$$

The last equality uses the zero row sum of H. The column mean has the same cancellation, using $\sum_i H_{ij} = 0$.

In Equation (4.29), the squared centred pointwise MI, $V(P_t)$, the conditional means and $I(P_t)$ all begin at order t^2. Only the unsquared pointwise-MI term in the bracket has a first-order contribution. Therefore Equation (4.29) supplies the first-order term:

$$g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.9}$$

Since g_{P_t} has mean zero, its variance is its second moment:

$$\begin{aligned} \tau^2(P_t) &= \sum_{i,j} (q_{ij} + tH_{ij}) \left\{ \frac{2tH_{ij}}{q_{ij}} + O(t^2) \right\}^2 \\ &= 4t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \end{aligned} \tag{A.10}$$

Comparing these three leading terms gives $V(P_t) \approx 2I(P_t)$ and $\tau^2(P_t) \approx 4V(P_t)$ for small positive MI along this path. At $t = 0$, the first-order sampling variance is zero; Section A.4 treats that case separately.

A.3 Dependence between the MI and variance estimates

Both $\widehat{I}(P)$ and $\widehat{V}(P)$ are calculated from the same sample. Under fixed positive support, their first-order covariance is

$$\mathrm{Cov}\{\widehat{I}(P), \widehat{V}(P)\} \approx \frac{1}{n_P} \mathrm{E}_P[\{\ell_P(X,Y) - I(P)\} g_P(X,Y)]\,. \tag{A.11}$$

The corresponding expression for Q replaces P and n_P with Q and n_Q. To see its direction near independence, use the fixed-margin path $P_t = (q_{ij} + tH_{ij})$ from Section A.2. For small positive t, the expansions there give

$$\ell_{P_t}(i,j) - I(P_t) = \frac{tH_{ij}}{q_{ij}} + O(t^2), \qquad g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.12}$$

Substituting both expressions into the first-order covariance above gives

$$\mathrm{Cov}\{\widehat{I}(P_t), \widehat{V}(P_t)\} \approx \frac{2t^2}{n_P} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O\left(\frac{t^3}{n_P}\right) \tag{A.13}$$

$$= \frac{2V(P_t)}{n_P} + O\left(\frac{t^3}{n_P}\right). \tag{A.14}$$

The first-order covariance is therefore positive for small positive MI along this path: samples with a larger MI estimate also tend to have a larger variance estimate. The two-sample rejection rate depends on both populations and the ratio of the difference to its standard error. At exact independence, $V(P_0) = 0$ and a second-order sampling approximation is needed.

A.4 The independence boundary

At independence, $p_{ij} = p_{i+}p_{+j}$, so every population pointwise-MI value is $\ell_P(i,j) = \log 1 = 0$. Consequently, $I(P) = V(P) = 0$, and the first-order change in MI is zero. Sampling can still produce positive estimated MI, but its leading variation comes from second-order cell errors.

To calculate this variation, fix a strictly positive independent table P and let $h = (h_{ij})$ describe a redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define

$$P_\varepsilon = P + \varepsilon h, \qquad f(\varepsilon) = I(P_\varepsilon), \tag{A.15}$$

for ε small enough that every cell remains positive. The expansion point is $\varepsilon = 0$, where $f(0) = I(P) = 0$. The second-order Taylor formula gives

$$f(\varepsilon) = f(0) + f'(0)\varepsilon + \frac{f''(0)}{2!}\varepsilon^2 + o(\varepsilon^2). \tag{A.16}$$

The first derivative is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = -\sum_{i,j} h_{ij} = 0. \tag{A.17}$$

Here the pointwise-MI terms vanish at independence, and the remaining terms cancel because the probabilities must still sum to one.

The row and column changes are $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Applying $\mathrm{d}^2(u \log u)/\mathrm{d}u^2 = 1/u$ to the joint and marginal terms in Equation (2.11) gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \tag{A.18}$$

Thus the leading change is

$$I(P_\varepsilon) = \frac{\varepsilon^2}{2} \left\{ \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}} \right\} + o(\varepsilon^2). \tag{A.19}$$

The first sum measures squared cell changes. The last two remove the contributions due to changed margins, leaving the change in association. The remainder is negligible relative to ε^2 as ε approaches zero.

For a multinomial sample, take $\varepsilon h = \widehat{P} - P$. Cell-probability errors typically have size proportional to $n^{-1/2}$. Squaring them makes $\widehat{I}(P)$ proportional to n^{-1} under independence. Under fixed table dimensions and positive cell probabilities, the likelihood-ratio result is

$$2n\widehat{I}(P) \xrightarrow{d} \chi^2_{(r-1)(c-1)} \tag{A.20}$$

(Wilks, 1938; Brillinger, 2004). This is the usual one-table likelihood-ratio, or G, test of independence. Its chi-squared distribution comes from quadratic cell errors, whereas Normal Wald and Expanded Welch use the first-order MI error. The two-sample first-order approximation therefore does not apply unchanged at exact independence.

A.5 The combined formula when a component is zero

For positive component degrees of freedom, substituting Equation (4.39) and its Q analogue into Equation (4.40) gives the equivalent combined form

$$\widehat{\nu}_{\text{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\widehat{\tau}^2(P)/(2n_P^3) + \widehat{\tau}^2(Q)/(2n_Q^3)}. \tag{A.21}$$

This direct form remains defined when one table has $\widehat{V} = \widehat{\tau}^2 = 0$ and the other gives a positive denominator. The zero contribution drops out, leaving the other table's component degrees of freedom. The implementation nevertheless requires both components to have positive, finite degrees of freedom, so it records this case as invalid.

Numerical derivative checks.

The analytic derivatives in Equations (4.18) and (4.29) were also checked by centred finite differences along the one-cell perturbation paths in Chapter 4. Small positive and negative steps were chosen to keep every cell positive; the change in MI or variance divided by the distance between the two steps agreed with its analytic derivative.

Appendix B

Experimental Details

This appendix specifies the population formulas, target-MI search, complete parameter grids, sampling procedure and outcome calculations. Together with the test formulas in Chapters 3 and 4, these details define the statistical experiment.

B.1 Copying construction and target selection

For a positive marginal vector a, the joint distribution is

$$p_{ij}(\lambda) = (1-\lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}. \tag{B.1}$$

Summing a row gives

$$\sum_j p_{ij}(\lambda) = (1-\lambda)a_i \sum_j a_j + \lambda a_i = a_i. \tag{B.2}$$

Summing a column similarly gives a_j. The copying branch contributes only to its matching category, while the independent branch contributes across every category. Thus the construction preserves both margins and has positive cells whenever $0 \leq \lambda < 1$.

MI increases continuously from zero to the entropy of a. With the margins fixed, differentiating the joint terms gives

$$\frac{\mathrm{d}^2 I\{P(\lambda)\}}{\mathrm{d}\lambda^2} = \sum_{i,j} \frac{(a_i \mathbf{1}_{\{i=j\}} - a_i a_j)^2}{p_{ij}(\lambda)} > 0. \tag{B.3}$$

The first derivative at zero is zero, because the change in probabilities sums to zero and all pointwise MI values are zero there. The positive second derivative makes

the first derivative positive for $\lambda > 0$. At the endpoint, $Y = X$, so

$$I\{P(1)\} = -\sum_i a_i \log a_i. \tag{B.4}$$

These facts give a unique copying probability for each positive target below the entropy endpoint.

For each population, calculate this endpoint and solve $I\{P(\lambda)\} - I_{\text{target}} = 0$ on $[0, 1]$ using Brent's method (Brent, 1973). Absolute and relative root tolerances are 10^{-15} and 10^{-14}. The endpoint brackets the search; sampled populations use the interior solution. Accept a table only when every cell is positive, its total and marginal errors are at most 10^{-12}, and its MI error is at most $10^{-12} + 10^{-8} I_{\text{target}}$. The maximum target in the study is 0.22 nats, below the smallest chosen entropy endpoint, approximately 0.5004 nats.

B.2 Four-cell supporting construction

For the supporting 4×4 comparison, begin with $B_{ij} = a_i a_j$ and transfer an amount t using

$$R(t) = aa^{\mathsf{T}} + t \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{B.5}$$

Every affected row and column gains and loses t, preserving its total. The decreasing cells have starting probability $a_1 a_2$, so

$$0 \le t < t_{\max} = a_1 a_2. \tag{B.6}$$

Solve $I\{R(t)\} = I_{\text{target}}$ by the same root search on $[0, t_{\max}]$. The target must lie below the endpoint MI, and the constructed table passes the same checks as the copying table.

In a binary table the copying construction is exactly this transfer:

$$P(\lambda) = (1 - \lambda) \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix} + \lambda \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix} \tag{B.7}$$

$$= \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix} + \lambda a_1 a_2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{B.8}$$

The second equality uses $a_1 + a_2 = 1$, giving $a_1 - a_1^2 = a_1 a_2$ and $a_2 - a_2^2 = a_1 a_2$. Hence $t = \lambda a_1 a_2$. This equivalence does not extend to the larger tables: copying changes cells throughout the table, whereas Equation (B.5) changes only four.

B.3 Complete parameter grids and sampling

Table sizes are $k \in \{2, 4, 8, 16\}$. Each population has identical row and column margins. Uniform margins give probability $1/k$ to every category; same skew uses $m_k(0.8)$ in both populations; different skew uses $m_k(0.7)$ in P and $m_k(0.8)$ in Q, with m_k defined by Equation (5.3). The same copying formula and root search apply to every primary-study setting.

Table B.1: Exact parameter sets; MI values are in nats.

Study	Settings
Equal MI	All four sizes and three margin regimes; $I(P) = I(Q) = 0.02$; equal sample sizes $n \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}$.
Association	4×4, different skew; $I(P) = I(Q) \in \{0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05\}$; $n \in \{100, 1000\}$.
Detection	All four sizes with different skew, plus uniform and same-skew 4×4; $I(P) = 0.02$, $I(Q) = 0.02 + \lvert\Delta\rvert$; $\lvert\Delta\rvert \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$; $n \in \{1000, 10000\}$.
Four-cell check	4×4, different skew; $(I(P), I(Q)) \in \{(0.02, 0.02), (0.02, 0.03)\}$; $n \in \{1000, 10000\}$; construction in Equation (B.5).

For one repetition, draw

$$(N_{P,11}, \ldots, N_{P,kk}) \sim \text{Multinomial}(n_P; p_{11}, \ldots, p_{kk}), \tag{B.9}$$

and independently draw the corresponding Q counts. Reshape these counts into their $k \times k$ tables and apply both methods to that pair. Repeat independently 20,000 times per distinct setting. The simulations use the PCG64 pseudo-random generator with master seed 20261006 and separate streams for the populations and settings. Repeated settings use one set of repetitions in every relevant display.

All samples retain their declared table dimensions, including empty rows and columns. Use natural logarithms, $0 \log 0 = 0$, and no smoothing. The correction for each plug-in MI is $(k-1)^2/(2n)$; do not truncate negative corrected estimates. Since $n_P = n_Q$, the two corrections cancel in their difference. Calculate the standard error using Equation (3.9) and the component and combined degrees of freedom using Chapter 4.

B.4 Validity and sparse-sample measures

Normal Wald requires a finite corrected difference, a finite positive standard error, $\widehat{V}(P)+\widehat{V}(Q) > 10^{-14}$, and a finite p-value. The threshold treats variances extremely close to zero as numerically unusable. Expanded Welch additionally requires positive finite degrees of freedom for both components and their combination, and a finite expanded p-value. A failed component makes Expanded Welch invalid under this rule, including the zero-component case in Section A.5.

The primary rate divides valid rejections by all repetitions. A method-valid rate uses only that method's valid repetitions; a common-valid rate uses only pairs where both methods are valid. Report the denominator for either conditional rate, which is undefined when the denominator is zero. Invalid outputs contribute no rejection to the primary rate.

Record the number of zero-count cells, empty rows and empty columns separately for each population. These are sampling diagnostics rather than eligibility rules. The minimum expected cell count is $\min_{ij} np_{ij}$, with the corresponding definition for Q. The study applies no minimum expected-count filter.

B.5 Bias, variance and degrees-of-freedom diagnostics

Let $\widehat{\Delta}_s$ be the corrected difference in repetition s. Its bias estimate is the average finite difference minus the known Δ. Its observed sampling variance is

$$\frac{1}{M-1}\sum_{s=1}^{M}\left(\widehat{\Delta}_s - \frac{1}{M}\sum_{u=1}^{M}\widehat{\Delta}_u\right)^2, \tag{B.10}$$

where M counts the finite differences used. Centre at their sample mean, rather than the true difference, so this measures variance separately from bias. Include

finite differences even when the test output is invalid. Individual corrected MI biases are calculated in the same way against $I(P)$ and $I(Q)$.

For the estimated/observed variance ratio, restrict both averages to the same repetitions with a finite difference and finite squared standard error. Retain finite zero variance estimates. Divide the mean squared standard error by the observed variance on this matched subset; the ratio is undefined if the observed variance is zero. Report its included fraction. The population comparison instead uses $V(P)/n_P + V(Q)/n_Q$ and the primary observed variance.

The Monte Carlo standard error of estimated bias is the observed SD divided by $\sqrt{M}$. Degrees-of-freedom medians and percentiles use valid Expanded Welch repetitions, with their included fraction reported. Percentile ranges describe the spread of those outputs, not confidence intervals for their median. These summaries expose estimation problems but do not isolate the causal contribution of each problem to rejection.

B.6 Simulation uncertainty

For a rate estimate $\widehat{r}$ from M independent repetitions, its approximate Monte Carlo standard error is $\{\widehat{r}(1-\widehat{r})/M\}^{1/2}$. Wilson intervals use $z = \Phi^{-1}(0.975)$ and have centre and half-width

$$\frac{\widehat{r} + z^2/(2M)}{1 + z^2/M}, \qquad \frac{z\sqrt{\widehat{r}(1-\widehat{r})/M + z^2/(4M^2)}}{1 + z^2/M}. \tag{B.11}$$

These intervals are used for both rejection and validity rates (Wilson, 1927).

For the paired method contrast, let D_s equal the Wald valid-rejection indicator minus the Expanded Welch valid-rejection indicator on pair s. The approximate interval for their average $\bar{D}$ is

$$\bar{D} \;\pm\; 1.96 \left\{ \frac{1}{M(M-1)} \sum_{s=1}^{M} (D_s - \bar{D})^2 \right\}^{1/2}. \tag{B.12}$$

Use the same common-valid subset when reporting a conditional paired contrast. Cross-construction comparisons use independent simulation streams, so their difference uncertainty is not calculated from paired indicators. All intervals describe simulation uncertainty at fixed populations, not simultaneous coverage over every setting.

Appendix C

Additional Results

These comparisons complete the controlled studies in Chapter 6. Each statistical setting uses 20,000 sampled table pairs and the two-sided 5% threshold.

C.1 Complete equal-MI results

Table C.1 reports every combination of four table sizes, three margin regimes and ten sample sizes from the equal-MI study. Both population MI values are 0.02 nats. Rejection rates divide valid rejections by all repetitions, while the final two columns report valid-result frequencies for Wald and Expanded Welch. The margin vectors are those in Table 5.2.

Table C.1: False-positive and valid-result rates for all 120 null settings.

Size	Margins	n	Wald	Expanded	Valid W	Valid E
2×2	Uniform	10	0.0590	0.0255	0.9907	0.9816
2×2	Uniform	20	0.0276	0.0181	0.9985	0.9852
2×2	Uniform	50	0.0081	0.0043	0.9999	0.9930
2×2	Uniform	100	0.0128	0.0054	1.0000	0.9989
2×2	Uniform	250	0.0302	0.0179	1.0000	1.0000
2×2	Uniform	500	0.0387	0.0276	1.0000	1.0000
2×2	Uniform	1,000	0.0445	0.0389	1.0000	1.0000
2×2	Uniform	2,500	0.0490	0.0464	1.0000	1.0000
2×2	Uniform	10,000	0.0465	0.0458	1.0000	1.0000
2×2	Uniform	50,000	0.0491	0.0490	1.0000	1.0000
2×2	Same skew	10	0.0358	0.0094	0.9571	0.6471

Size	Margins	n	Wald	Expanded	Valid W	Valid E
2×2	Same skew	20	0.0238	0.0146	0.9976	0.9542
2×2	Same skew	50	0.0095	0.0047	1.0000	0.9999
2×2	Same skew	100	0.0114	0.0040	1.0000	1.0000
2×2	Same skew	250	0.0263	0.0133	1.0000	1.0000
2×2	Same skew	500	0.0390	0.0255	1.0000	1.0000
2×2	Same skew	1,000	0.0452	0.0372	1.0000	1.0000
2×2	Same skew	2,500	0.0483	0.0456	1.0000	1.0000
2×2	Same skew	10,000	0.0500	0.0490	1.0000	1.0000
2×2	Same skew	50,000	0.0495	0.0493	1.0000	1.0000
2×2	Different skew	10	0.0407	0.0115	0.9816	0.7557
2×2	Different skew	20	0.0271	0.0163	0.9980	0.9738
2×2	Different skew	50	0.0086	0.0044	1.0000	1.0000
2×2	Different skew	100	0.0106	0.0047	1.0000	1.0000
2×2	Different skew	250	0.0273	0.0137	1.0000	1.0000
2×2	Different skew	500	0.0404	0.0278	1.0000	1.0000
2×2	Different skew	1,000	0.0425	0.0360	1.0000	1.0000
2×2	Different skew	2,500	0.0496	0.0466	1.0000	1.0000
2×2	Different skew	10,000	0.0504	0.0495	1.0000	1.0000
2×2	Different skew	50,000	0.0495	0.0493	1.0000	1.0000
4×4	Uniform	10	0.0985	0.0512	1.0000	0.9995
4×4	Uniform	20	0.0813	0.0470	1.0000	1.0000
4×4	Uniform	50	0.0472	0.0308	1.0000	1.0000
4×4	Uniform	100	0.0210	0.0143	1.0000	1.0000
4×4	Uniform	250	0.0217	0.0154	1.0000	1.0000
4×4	Uniform	500	0.0283	0.0217	1.0000	1.0000
4×4	Uniform	1,000	0.0382	0.0331	1.0000	1.0000
4×4	Uniform	2,500	0.0442	0.0420	1.0000	1.0000
4×4	Uniform	10,000	0.0467	0.0458	1.0000	1.0000
4×4	Uniform	50,000	0.0490	0.0490	1.0000	1.0000
4×4	Same skew	10	0.0798	0.0294	0.9606	0.6637
4×4	Same skew	20	0.0478	0.0308	0.9996	0.9547
4×4	Same skew	50	0.0141	0.0075	1.0000	0.9999
4×4	Same skew	100	0.0068	0.0037	1.0000	1.0000

Size	Margins	n	Wald	Expanded	Valid W	Valid E
4×4	Same skew	250	0.0192	0.0084	1.0000	1.0000
4×4	Same skew	500	0.0325	0.0208	1.0000	1.0000
4×4	Same skew	1,000	0.0392	0.0319	1.0000	1.0000
4×4	Same skew	2,500	0.0437	0.0400	1.0000	1.0000
4×4	Same skew	10,000	0.0481	0.0472	1.0000	1.0000
4×4	Same skew	50,000	0.0508	0.0505	1.0000	1.0000
4×4	Different skew	10	0.1332	0.0493	0.9909	0.7776
4×4	Different skew	20	0.0800	0.0444	1.0000	0.9778
4×4	Different skew	50	0.0240	0.0118	1.0000	0.9999
4×4	Different skew	100	0.0124	0.0059	1.0000	1.0000
4×4	Different skew	250	0.0238	0.0119	1.0000	1.0000
4×4	Different skew	500	0.0331	0.0225	1.0000	1.0000
4×4	Different skew	1,000	0.0394	0.0331	1.0000	1.0000
4×4	Different skew	2,500	0.0427	0.0398	1.0000	1.0000
4×4	Different skew	10,000	0.0481	0.0473	1.0000	1.0000
4×4	Different skew	50,000	0.0476	0.0471	1.0000	1.0000
8×8	Uniform	10	0.1397	0.0799	1.0000	0.9996
8×8	Uniform	20	0.0703	0.0584	1.0000	1.0000
8×8	Uniform	50	0.0688	0.0586	1.0000	1.0000
8×8	Uniform	100	0.0736	0.0651	1.0000	1.0000
8×8	Uniform	250	0.0362	0.0328	1.0000	1.0000
8×8	Uniform	500	0.0222	0.0207	1.0000	1.0000
8×8	Uniform	1,000	0.0266	0.0245	1.0000	1.0000
8×8	Uniform	2,500	0.0333	0.0318	1.0000	1.0000
8×8	Uniform	10,000	0.0431	0.0426	1.0000	1.0000
8×8	Uniform	50,000	0.0498	0.0495	1.0000	1.0000
8×8	Same skew	10	0.1053	0.0546	0.9606	0.6865
8×8	Same skew	20	0.0941	0.0740	0.9993	0.9542
8×8	Same skew	50	0.0503	0.0360	1.0000	1.0000
8×8	Same skew	100	0.0189	0.0121	1.0000	1.0000
8×8	Same skew	250	0.0067	0.0045	1.0000	1.0000
8×8	Same skew	500	0.0084	0.0060	1.0000	1.0000
8×8	Same skew	1,000	0.0242	0.0198	1.0000	1.0000

Size	Margins	n	Wald	Expanded	Valid W	Valid E
8×8	Same skew	2,500	0.0414	0.0388	1.0000	1.0000
8×8	Same skew	10,000	0.0456	0.0444	1.0000	1.0000
8×8	Same skew	50,000	0.0503	0.0499	1.0000	1.0000
8×8	Different skew	10	0.1938	0.1072	0.9889	0.7955
8×8	Different skew	20	0.1805	0.1434	1.0000	0.9768
8×8	Different skew	50	0.1218	0.0956	1.0000	1.0000
8×8	Different skew	100	0.0706	0.0517	1.0000	1.0000
8×8	Different skew	250	0.0384	0.0289	1.0000	1.0000
8×8	Different skew	500	0.0276	0.0217	1.0000	1.0000
8×8	Different skew	1,000	0.0311	0.0267	1.0000	1.0000
8×8	Different skew	2,500	0.0411	0.0384	1.0000	1.0000
8×8	Different skew	10,000	0.0467	0.0454	1.0000	1.0000
8×8	Different skew	50,000	0.0486	0.0483	1.0000	1.0000
16×16	Uniform	10	0.1764	0.0454	1.0000	0.9960
16×16	Uniform	20	0.1051	0.0822	1.0000	1.0000
16×16	Uniform	50	0.0459	0.0425	1.0000	1.0000
16×16	Uniform	100	0.0522	0.0485	1.0000	1.0000
16×16	Uniform	250	0.0858	0.0811	1.0000	1.0000
16×16	Uniform	500	0.0655	0.0635	1.0000	1.0000
16×16	Uniform	1,000	0.0339	0.0331	1.0000	1.0000
16×16	Uniform	2,500	0.0227	0.0222	1.0000	1.0000
16×16	Uniform	10,000	0.0348	0.0344	1.0000	1.0000
16×16	Uniform	50,000	0.0469	0.0469	1.0000	1.0000
16×16	Same skew	10	0.1099	0.0630	0.9580	0.6643
16×16	Same skew	20	0.1179	0.0986	0.9995	0.9546
16×16	Same skew	50	0.0887	0.0732	1.0000	1.0000
16×16	Same skew	100	0.0537	0.0442	1.0000	1.0000
16×16	Same skew	250	0.0190	0.0155	1.0000	1.0000
16×16	Same skew	500	0.0086	0.0074	1.0000	1.0000
16×16	Same skew	1,000	0.0065	0.0059	1.0000	1.0000
16×16	Same skew	2,500	0.0143	0.0134	1.0000	1.0000
16×16	Same skew	10,000	0.0449	0.0440	1.0000	1.0000
16×16	Same skew	50,000	0.0464	0.0462	1.0000	1.0000

Size	Margins	n	Wald	Expanded	Valid W	Valid E
16×16	Different skew	10	0.2063	0.1258	0.9899	0.7827
16×16	Different skew	20	0.2301	0.2007	1.0000	0.9758
16×16	Different skew	50	0.2200	0.2016	1.0000	1.0000
16×16	Different skew	100	0.2155	0.1984	1.0000	1.0000
16×16	Different skew	250	0.2352	0.2180	1.0000	1.0000
16×16	Different skew	500	0.2700	0.2550	1.0000	1.0000
16×16	Different skew	1,000	0.2522	0.2402	1.0000	1.0000
16×16	Different skew	2,500	0.0925	0.0882	1.0000	1.0000
16×16	Different skew	10,000	0.0462	0.0452	1.0000	1.0000
16×16	Different skew	50,000	0.0477	0.0474	1.0000	1.0000

Figure C.1 supplies the four size–margin curves not displayed in the main null figures: uniform and same-skew 2×2 and 8×8 populations. Error bars are pointwise Wilson 95% intervals; the lower strips show validity.

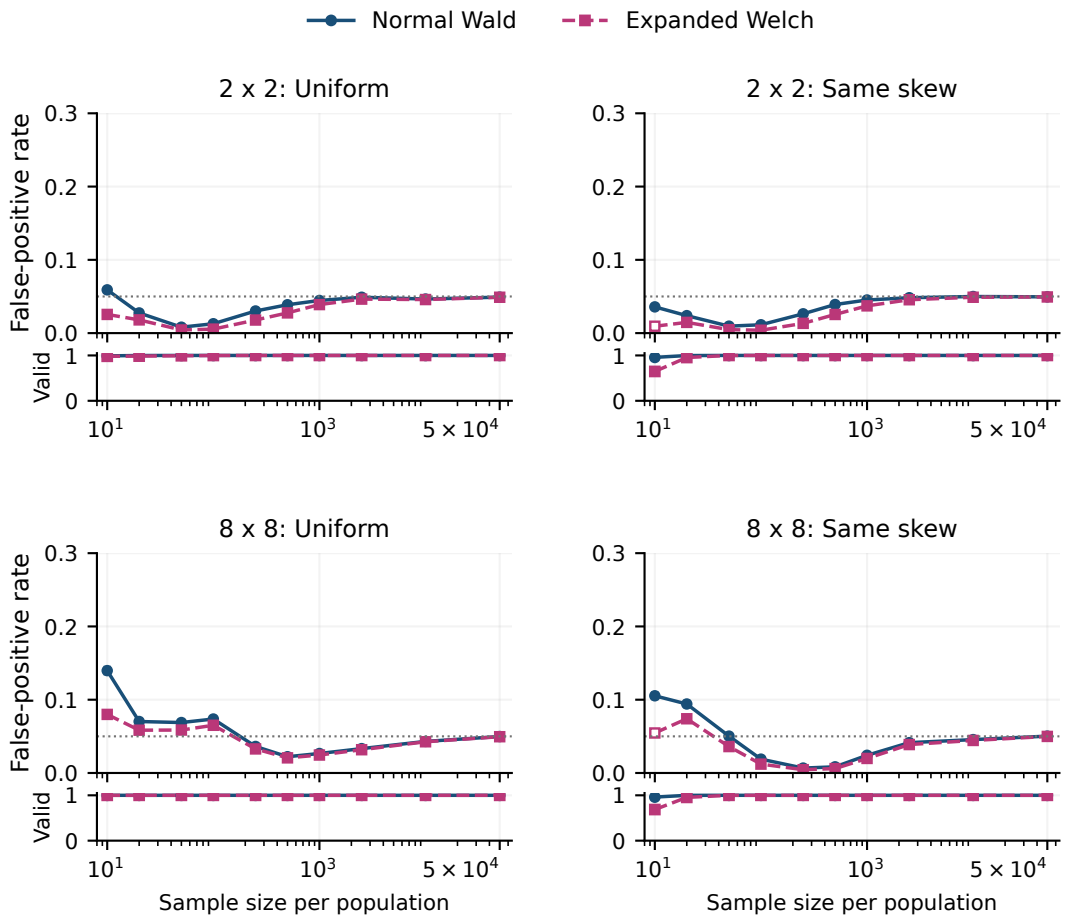


Figure C.1: Additional equal-MI comparisons under uniform and same-skew margins.

The smaller tables again give low false-positive rates at intermediate sample sizes. At $n = 100$, Wald rejects at 0.0128 under uniform margins and 0.0114 under same skew, with still lower Expanded Welch rates. In 8×8 tables, uniform margins instead give rates 0.0737 and 0.0651 at this size, whereas same-skew margins give 0.0189 and 0.0121. The effect of the margin regime therefore changes with table size. By $n = 50{,}000$, all four comparisons have rates between 0.0489 and 0.0503 and full validity.

C.2 Supporting estimation diagnostics

Table C.2 gives the diagnostics behind the association-strength figure. Est./obs. is the ratio of mean squared standard error to observed sampling variance. Pop./obs. replaces the estimated variance with the population first-order variance $V(P)/n + V(Q)/n$. DF denotes combined degrees of freedom; its bracketed range gives the 5th and 95th percentiles. All repetitions have finite diagnostics and valid test outputs in these settings.

Table C.2: Diagnostics for 4×4 different-skew populations with equal MI.

n	MI	Bias	DF median [5th, 95th]	Est./obs.	Pop./obs.
100	0.0001	0.00751	4.23 [2.25, 8.77]	1.919	0.007
100	0.0003	0.00724	4.33 [2.27, 9.11]	1.850	0.019
100	0.001	0.00666	4.53 [2.32, 9.62]	1.851	0.062
100	0.003	0.00606	4.98 [2.45, 10.63]	1.760	0.166
100	0.01	0.00567	6.35 [2.92, 13.61]	1.611	0.416
100	0.02	0.00569	8.10 [3.65, 17.17]	1.432	0.608
100	0.05	0.00588	13.80 [6.32, 27.97]	1.176	0.808
1,000	0.0001	-0.00011	8.62 [4.75, 14.47]	1.736	0.040
1,000	0.0003	-0.00018	8.79 [4.86, 14.52]	1.733	0.114
1,000	0.001	-0.00017	9.62 [5.44, 15.50]	1.590	0.305
1,000	0.003	-0.00015	12.32 [7.21, 19.21]	1.377	0.579
1,000	0.01	-0.00022	21.93 [14.18, 31.82]	1.163	0.835
1,000	0.02	-0.00012	35.70 [24.60, 49.15]	1.061	0.895
1,000	0.05	-0.00003	78.24 [59.53, 100.65]	1.004	0.948

At the smallest MI targets, population first-order variance accounts for only a small fraction of the observed spread at either sample size. Meanwhile, the plug-in

variance estimate exceeds that spread. Using estimated rather than population inputs therefore changes the error as well as its size. Both comparisons become much closer to one as association strengthens, especially at $n = 1{,}000$. This supports reading the low rejection rates together with estimation error, rather than attributing them solely to the Student t distribution.

Table C.3 continues the main diagnostic comparison to $n = 50{,}000$ at common MI 0.02. All outputs are valid. At 4×4 the difference bias has Monte Carlo standard error approximately 0.000010 nats; at 16×16 it is approximately 0.000012. The biases are small, the estimated/observed variance ratios are near one, and the large degrees of freedom bring the two reference distributions close together. The remaining population/observed discrepancy at 16×16 is larger than that at 4×4.

Table C.3: Different-skew null diagnostics at $n = 50{,}000$.

Size	Bias	Est./obs.	Pop./obs.	DF median
4×4	-0.00002	1.006	1.002	1436.59
16×16	-0.00002	1.017	0.951	1146.78

C.3 Detection over the full difference range

Figures C.2 and C.3 show the same settings as the main detection figures, extending the horizontal range through 0.2 nats. They retain the zero-difference points and Wilson interval bands. The population targets are $I(P) = 0.02$ and $I(Q) = 0.02+|\Delta|$; every result is valid in these settings.

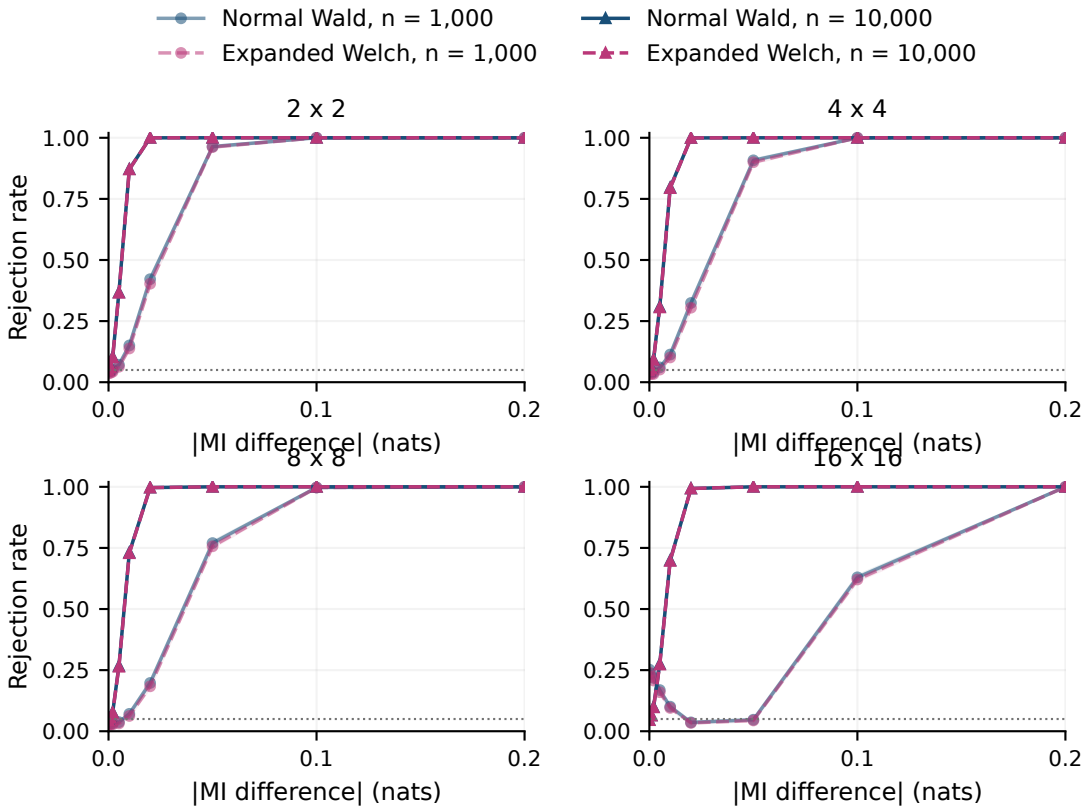


Figure C.2: Full-range detection under different-skew margins.

At $n = 1{,}000$, detection of a 0.05-nat difference falls with table size: Wald rates are 0.9642, 0.9079, 0.7699 and 0.0471 from 2×2 through 16×16. The last curve first falls from its high null rate, then rises to 0.6297 at 0.1 nats and 0.9997 at 0.2. Unequal bias opposes the small negative true differences there, as discussed in the main results. At $n = 10{,}000$, both methods detect every simulated 0.05-nat difference at all four sizes.

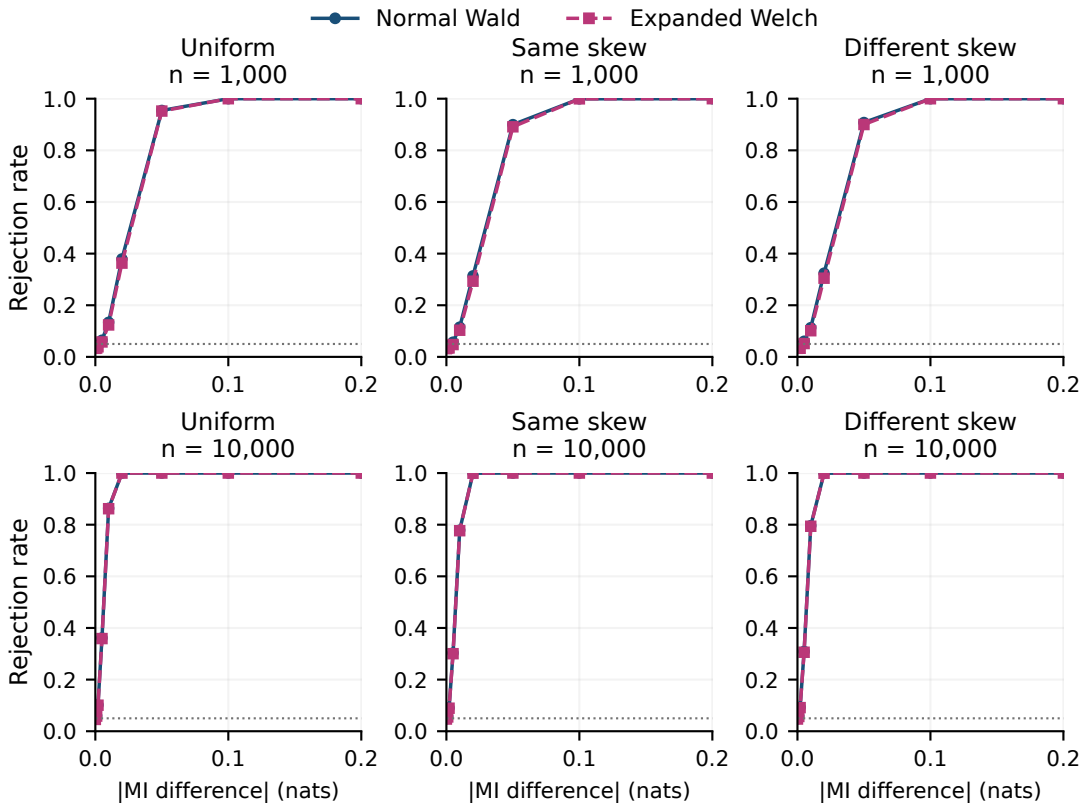


Figure C.3: Full-range detection across 4×4 margin regimes.

The margin regimes differ most at intermediate effects, before rejection approaches one. At 0.2 nats, both methods reject every simulated sample at both sample sizes in all three regimes. Near this upper limit, a smaller difference between methods says little about their relative performance for weaker effects; the enlarged main figures show those effects more clearly.

C.4 Four-cell construction check

Table C.4 compares the copying and four-cell populations at matched 4×4 different-skew margins and MI targets. All outputs and diagnostics are finite and valid. The sampled populations differ between constructions, so the cross-construction comparison uses independent repetitions. Within each construction, Wald and Expanded Welch receive the same table pairs.

Table C.4: Rejection and difference bias under two population constructions.

Construction	n	$\lvert\Delta\rvert$	Wald	Expanded	Bias
Copying	1,000	-0	0.0394	0.0331	-0.00012
Copying	1,000	0.01	0.1132	0.1013	-0.00004
Copying	10,000	-0	0.0481	0.0473	-0.00001
Copying	10,000	0.01	0.7962	0.7940	0.00002
Four-cell	1,000	-0	0.0403	0.0345	-0.00018
Four-cell	1,000	0.01	0.1148	0.1032	-0.00018
Four-cell	10,000	-0	0.0502	0.0493	-0.00001
Four-cell	10,000	0.01	0.7864	0.7844	-0.00003

The same qualitative method comparison persists: at $n = 1{,}000$, Expanded Welch lowers an already-low null rate and reduces detection; at 10,000 its effect is smaller. The construction still changes the numerical detection rate. For the 0.01-nat difference at $n = 10{,}000$, Wald detects at 0.7962 under copying and 0.7865 under the four-cell transfer. The difference, 0.0098, has approximate independent-stream 95% interval [0.0018, 0.0177]. Thus the check supports the direction of the method comparison at these settings, not identical performance under every association pattern.

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